

Fiscal Narratives and Inflation^{*}

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— draft —

Abstract

This paper investigates how media narratives on fiscal policy shape household’s inflation expectations. We collect a large corpus of newspaper articles reporting on fiscal policy from four major German newspapers spanning from 2006 to March 2025. Using a large language model (ChatGPT) we introduce a strategy to automatically identify different fiscal narratives in text and construct narrative indices out of this data. We then estimate the effect of these narrative indices on household inflation expectations and find that they all have a positive significant effect varying in size. Lastly, we measure how fiscal narratives affect the transmission of a government spending shock to the economy and find that some of the narratives have an amplifying effect while others dampen the impact.

Keywords: fiscal policy, narratives, government spending shock, ChatGPT

JEL classification: E31, E62, E37

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1 Introduction

The recent surge in inflation in the US and Europe that began in 2021 was accompanied by a high debt level due to deficit-financed fiscal interventions that were introduced to cushion the economic impact of the COVID-19 pandemic. Due to these circumstances, the question on the relation between fiscal policy and inflation gained renewed interest in the literature. [Barro & Bianchi \(2024\)](#), for example, show empirically that in the aggregate of 17 euro area countries in 2020-2023, inflation responded positively to a composite government spending variable. Further, [Bassetto & Miller \(2025\)](#) build a model in which they show how inflation and fiscal deficits can suddenly become connected through an information channel, generating sudden inflation. In a similar spirit, [De Fiore et al. \(2024\)](#) estimate the effect of the announcement of three major US stimulus packages (the American Rescue Plan, the CARES Act, and the Tax Cuts and Jobs Act) and find that financially literate households adjust their inflation expectations upwards.

In this paper, we want to shed more light on the aforementioned information channel that seems to play a significant role in the effects of fiscal policy shocks. Unlike professional forecasters or financial market participants, households typically do not directly observe fiscal policy interventions, but instead rely on intermediaries such as the news media to make sense of economic developments. In most cases, journalists do not exclusively report on factual events, but also interpret, frame, and narrate fiscal policy in ways that can shape public beliefs about government action, debt, and inflation. The majority of households do not know about economic theories that predict the effects of a government spending shock or fiscal deficits. If they establish a link between e.g., fiscal expansion and inflation then most likely because they learned about it in the news.

So far, [Coibion et al. \(2021\)](#) and [Andrade et al. \(2025\)](#) investigate in survey experiments the link between information on the fiscal stance and inflation expectations. The former find for the US that only information on the change of the government debt level leads to an upwards revision of inflation expectations, while the latter observe for German households that only those who already believe the fiscal resources are constrained expect higher inflation when learning about a growing debt-to-GDP ratio. Again, the households' perceptions of the fiscal space might be strongly influenced by the media. As an illustration; if newspapers deem the fiscal space to be limited due to, in their view excessive debt levels, agents consuming these news might follow this view. Based on [Andrade et al. \(2025\)](#) this would imply that any announcement of future fiscal expansion measures would lead to an increase of inflation expectations in these households that might translate into actual inflation.

As an alternative to these information provision experiments, we want to directly

analyze the information set of households which we proxy with news articles. In this paper, we introduce a systematic approach using Artificial Intelligence (AI) to measure fiscal narratives in Germany. We want to answer the question whether media outlets establish a link between fiscal expansion and inflation in their reporting, as well as measure their narratives on government debt. Building on these fiscal narratives, we analyze how this type of information affects household inflation expectations and estimate their impact on the transmission of government spending shocks to output and inflation in Germany.

Our approach is as follows; we collect a set of newspaper articles that contain at least one of a list of terms related to fiscal policy to aim for only those that are of interest for our analysis. We focus on four major news outlets in Germany: *Die Bild*, *Frankfurter Allgemeine Zeitung*, *Die Süddeutsche Zeitung*, and *die tageszeitung*. These newspapers represent different types of media: tabloid press, conservative broadsheets and critical left-wing journalism. By analyzing this diverse selection, we aim to assess the differences in their reporting on fiscal policy, but also capture an as wide as possible range of public opinions.

Our findings provide empirical insights on the heterogeneity of news and the importance of news to the formation of macro expectations. In doing so, we also contribute to the development of models that incorporate heterogeneous beliefs. First, we evaluate the effect of the fiscal narratives on household inflation expectations using the ECB Consumer Expectations Survey and find that three out of four narrative indicators show significant effects on 12-month ahead household inflation expectations. Articles linking fiscal expansion to inflation raise household inflation expectations by 0.055 percentage points (pp), narratives framing government spending as desirable raise expectations by 0.018 pp, and calls for fiscal consolidation raise expectations by 0.011 pp. The effect of debt sustainability narratives is state-dependent: insignificant in the full sample but strongly significant in the post-2022 high-inflation regime ($\hat{\beta} = 0.148^{***}$), consistent with rational inattention (?) — households only process abstract fiscal sustainability concerns when inflation is already salient. We further document that narrative effects are heterogeneous across demographic groups: they are significantly stronger for low-education and middle-aged households, and women respond more strongly to the expansion→inflation and spending narratives. The results are robust to Double Machine Learning estimation, Post-Double Selection Lasso, alternative clustering schemes, subsample splits, and the exclusion of major fiscal announcement months.

In a second step, we investigate the non-linear effect of the fiscal narratives on the transmission of government spending shocks. News articles conveying the government should decrease its spending dampen the effect of a fiscal shock on output and inflation while the opposite holds true when they report the government should increase its spending. We combine these two narratives into a composite fiscal narrative index —

spending weighted positively and saving weighted negatively — which amplifies the effect of a government spending shock on both GDP and inflation when it takes high values. This way we can show that narratives not only affect subjective expectations, but also have an effect on macro variables.

Related Literature. Our paper is part of the expanding body of research that seeks to extract structured information from textual sources, such as sentiments or expectations. See, for example, [Bybee \(2023\)](#), [Shapiro et al. \(2022\)](#), [Picault et al. \(2022\)](#), [Ellingsen et al. \(2022\)](#) and [Angelico et al. \(2022\)](#). Another example is [Aruoba & Drechsel \(2024\)](#) who use textual information in documents that economists at the Federal Reserve prepare for Federal Open Market Committee (FOMC) meetings, to identify monetary policy shocks.

Recent studies apply natural language processing techniques likewise to investigate fiscal policy related questions. [Latifi et al. \(2024\)](#) use all the parliamentary speeches in the German Bundestag from 1960 to 2021 and derive fiscal sentiments (i.e., expansionary vs. contractionary fiscal policy stance) from it using machine learning techniques. Furthermore, they use these fiscal sentiments to show that changes in fiscal sentiment cause a shift in government spending. In an extension, [Tillmann \(2025\)](#) estimates the effect of a monetary policy shock on fiscal sentiments. He uses the same fiscal sentiments measure as in [Latifi et al. \(2024\)](#) and finds that monetary tightening causes a drop in fiscal sentiments. [Lieb et al. \(2025\)](#) use all the publicly available postwar communications of US presidents to predict the direction and size of future tax changes. The aforementioned studies all focus on the policy makers themselves, using their speeches to uncover their fiscal stance. In contrast, our paper puts the spotlight on those affected by fiscal policy, namely the public. By leveraging news articles we can learn about the public perception of fiscal policy and how it affects households’ expectations.

Since [Shiller \(2017\)](#) there has been an emerging interest in studying narratives from an economic perspective. In [Shiller \(2020\)](#), Shiller coins the definition of economic narratives as “*stories that offer interpretations of economic events, or morals, of hints of theories about the economy*”. In that vein, [Andre et al. \(2021\)](#) and [Andre et al. \(2024\)](#) follow a survey approach to measure economic narratives of households about inflation by analyzing open-ended survey questions in which participants explain what they think caused the latest surge in inflation. Building on this analysis, [Trebbs \(2024\)](#) identifies supply and demand narratives within newspaper articles on inflation. Going beyond broad economic narratives, we adopt Shiller’s definition as “*stories that offer interpretations of economic events*” and apply it specifically to the context of fiscal policy. In our setting, we understand fiscal narratives as the recurrent stories in the media that interpret, explain, or give meaning to fiscal policy actions and their economic consequences.

This paper is structured as follows: Section 2 describes our dataset and the method of measuring fiscal narratives in news texts, Section 3 then presents the resulting indices that we construct, afterwards in Section 4 we validate the indices. In Section 5, we estimate their effect on household inflation expectations, followed by analyzing how fiscal narratives affect the transmission of government spending shocks to GDP and inflation in Section 6. Finally, Section 7 concludes.

2 Data and measurement

2.1 Newspaper data

We compile a large dataset of daily news articles from four major German daily newspapers: *Frankfurter Allgemeine Zeitung* (FAZ), *Süddeutsche Zeitung* (SZ), *Bild*, and *die tageszeitung* (TAZ). Three of them (Bild, SZ and FAZ) are the most widespread daily newspapers in Germany and all of them appeal to a different segment of the German population. Bild is the most popular tabloid in Germany and counts 1,252,394 sold units of its Sunday edition in the fourth quarter of 2024 (source: IVW, 2025¹). Its style of reporting is characterized by exaggerations and the use of polarizing language. In contrast, FAZ and SZ are the most popular German quality newspapers which are also known to be more conservative in their views. Each of them sold 512,520 and 460,789 units, respectively, in 2024Q4 (source: IVW, 2025). Lastly, the TAZ stands for independent, critical and left-leaning journalism but has the comparably smallest number of sold units in 2024Q4, namely 44,765 (source: IVW, 2025). With this diverse range of media outlets we try to cover the major streams of news reporting to represent an as large as possible share of the news consuming German population.

Our sample spans from January 2006 for FAZ, SZ, and TAZ, from August 2011 for Bild, to March 2025. We either manually collect the articles from the publishing houses' online archives or web scrape them using the open-source platform mediacloud.org. In both cases we limit the number of articles we collect by applying a set of keywords associated with fiscal topics and consider only those that contain at least one of the keywords. The list of keywords comprises all words of the three fiscal dictionaries reported in Appendix A. With this procedure, we collect 514,940 articles in total. The amount of articles per news outlet are listed in Table 1.

Figure 1 depicts the total monthly distribution of articles by newspaper across time (the blue line). One can see that there is substantial time variation of how often the considered papers report on fiscal topics. For a first exploration of the data set we apply a dictionary of terms related to fiscal expansion and another one on fiscal expansion

¹Informationsgemeinschaft zur Feststellung der Verbreitung von Werbeträgern e.V. (IVW)

(see Appendix A) on the text corpus and filter those articles containing at least one of these terms. The results of this exercise are also shown in Figure 1. In general newspapers tend to report more on topics related to fiscal expansion (orange line), but especially in the past year, starting in 2024, fiscal consolidation (green line) becomes more prevalent in the news reporting.

	Bild	FAZ	TAZ	SZ
Sample Span	2011-2025	2006-2025	2006-2025	2006-2025
Total	47,230	58,290	77,783	331,637
Fiscal Expansion	21,476	27,852	36,461	154,422
Fiscal Consolidation	3,700	9,214	10,921	36,810

Table 1: Number of articles per media outlet.

The sample starts in January 2006, for FAZ, SZ and TAZ and in August 2011 for Bild. It ends in March 2025.

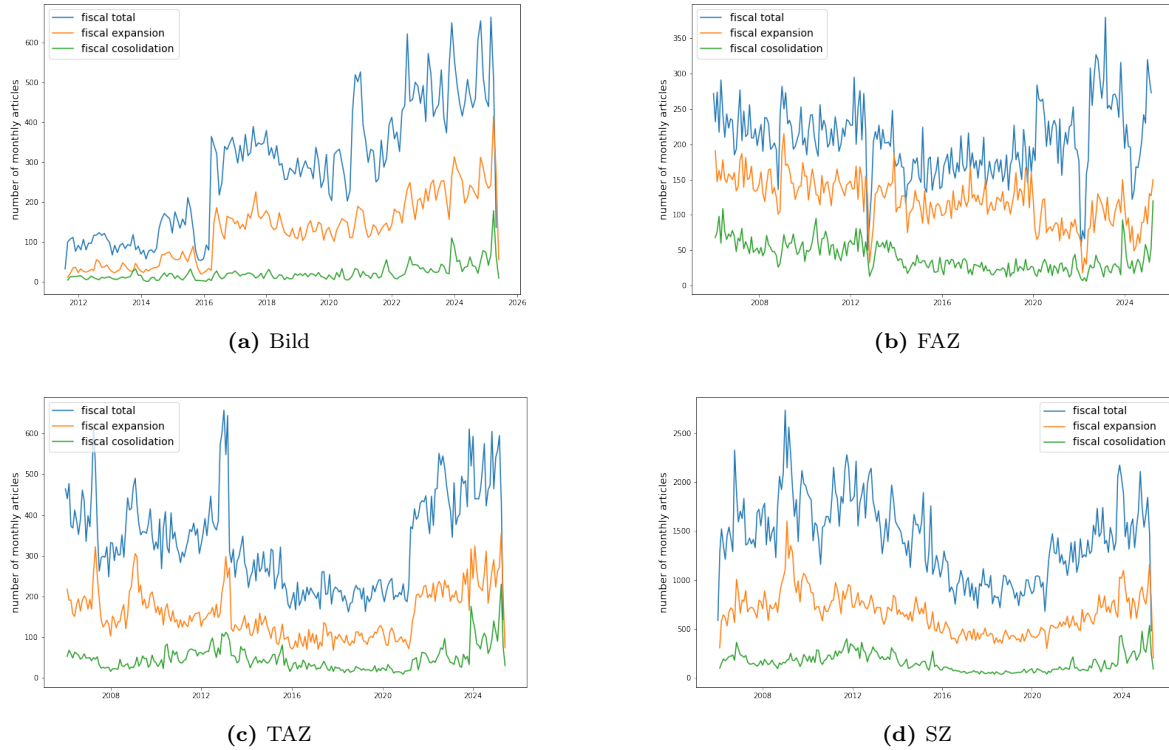


Figure 1: Number of monthly articles across time by newspaper.

This figure presents the number of monthly articles when filtering for keywords associated with fiscal expansion (orange), fiscal consolidation (green), or general fiscal topics (blue) for each of the media outlets Bild, FAZ, SZ, and TAZ.

2.2 Textual analysis of fiscal narratives

We conduct the main textual analysis with the large language model ChatGPT-4o-mini by OpenAI. Our goal is to measure the prevalence of certain fiscal narratives in

the media that allow for conclusions on the link between fiscal expansion and inflation. The narratives we want to measure are the following four:

1. Fiscal expansion will lead to inflation (referred to as *expansion leads to inflation* or *expansion $\Rightarrow$ inflation* narrative).
2. The German government debt level is unsustainably high (referred to as *unsustainable debt* narrative).
3. The German government should spend more and conduct fiscal expansion (referred to as *spending* narrative).
4. The German government should reduce its spending and conduct fiscal consolidation (referred to as *saving* narrative).

This choice of narratives is partly motivated by [Bassetto & Miller \(2025\)](#) who report the numbers of Google Searches of the term inflation in connection with fiscal policy related terms. They show that these searches especially rose starting from 2022. Further, they construct a model that is based on the assumption that at some point agents start to acquire more information on the government debt level. If more agents have this information, a transition into a regime in which expansion will generate inflation becomes more likely, in the case that the government is fiscally constraint². These model assumptions and predictions can be captured by narratives that on the one hand link inflation and fiscal expansion (i.e., narrative 1), and on the other hand by narratives that convey that the government is risking to become fiscally constraint (i.e., narratives 2 and 4). Lastly, we are also interested in the opposite case, i.e., the spending narrative.

To measure the four narratives we design three prompts with which we automatically classify the text corpus using the OpenAI batch application programming interface (API). In the first prompt, to identify the fiscal expansion leads to inflation narrative, we ask the model to determine whether the article establishes a narrative that fiscal expansion (e.g., government spending increases, tax cuts) will lead to inflation. In the second prompt to measure the unsustainable debt narrative, we ask the model: "Does the article suggest that government debt levels in Germany are unsustainable?" And in the third prompt we detect the spending/saving narratives by asking if the presented article conveys an opinion on whether the German government should save more or increase its spending. The exact prompts can be found in [Appendix B](#). For each prompt we receive a binary yes/no answer or the answers "save" and "spend" which we use to classify the articles according to the narratives.

To then create indicators out of the raw narrative classification counts, we create a monthly weighted sum by popularity of the news outlet. This way we account for

²Fiscal constraints can also be politically driven by for example limiting the possibility to increase taxes or to take on higher debt ([Bassetto & Miller, 2025](#)). In Germany such a political instrument limiting government borrowing and ensuring fiscal discipline is the so called "debt break", a constitutional rule introduced in 2009.

their differences in dissemination and their likelihood of being seen by the public. The most popular outlet in our news set is the Bild newspaper with approximately 7.35 million readers per daily edition. SZ, FAZ and TAZ each reach 1.3, 0.94, and 0.296 million readers per edition (source: ma 2022 Tageszeitungen³). These single indicators then help assess the public’s views and perceptions on fiscal policy and its relation to inflation.

3 Indices of fiscal narratives

We present the results of the newspaper article classification for each media outlet and narrative in Figure 2. Each graph shows the absolute monthly numbers of articles of the respective newspaper that got classified into one of the four narratives. The sample of Bild articles starts in August 2011 and the others in January 2006. One can see three different types of variation: there is substantial time variation within each narrative, but also across narratives, and there is heterogeneity between the different newspapers.

Comparing the number of articles classified into the unsustainable debt narrative across newspapers, one can see that there are similar peaks, even though their relative size differs. According to our measurement, this narrative was especially prevalent in the years of the Great Financial Crisis (GFC) and later in the years of the euro crisis with peaks in 2010 and 2012. In the mid 2010’s when the European sovereign debt was tackled, the unsustainable debt narrative lost importance, but gained popularity later on starting with the COVID-19 pandemic in 2020 with another surge in 2022, when Russia invaded Ukraine. Comparing the newspapers, one can see that for TAZ and SZ the peaks during the GFC and the invasion of Ukraine are similarly high, while the emphasis on this narrative in the FAZ was more pronounced during the GFC and euro crisis.

In the case of the spending vs. saving narratives, the differences between newspapers become even bigger. While the saving narrative seems to dominate the FAZ reporting during the GFC and the sovereign debt crisis, both are more on par in the TAZ and SZ with larger peaks in the spending narrative. In contrast, for them the prevalence of the two narratives diverges towards the end of the sample with the spending narrative becoming the leading narrative, especially for TAZ. For Bild, the spending narrative has its first bigger peak at the onset of the COVID-19 crisis, and after the invasion of Ukraine, both narratives surge and become equally important.

Lastly, also the distributions of the articles classified into the fiscal expansion leads to inflation narrative show a similar pattern, especially across Bild, FAZ, and TAZ with

³Numbers on newspaper readerships are obtained from Arbeitsgemeinschaft Media-Analyse e.V. (agma).

peaks in 2022. For all of these three media outlets, this type of narrative did not seem to play an important role in the years before. This picture is quite different in the case of SZ. Here, also during the GFC and euro crisis the prevalence of the fiscal expansion narrative leads to inflation narrative is similarly high compared to the end of the sample.



Figure 2: Narrative classification of articles by newspaper.
The sample periods vary by news source.

In Figure 3 we present the final fiscal narrative indices that we construct by weighting the individual newspaper results by their audience size. Overall one can see that most of the articles get classified into the spending narrative, directly followed by the saving narrative. Another general observation is that even though the amount of articles

related to fiscal policy was more or less stable across time with a small reduction the mid 2010's (see Figure 1), this drop is much stronger in the fiscal narrative indices. We interpret this as an indication that the narratives capture something beyond the bare attention to fiscal policy.

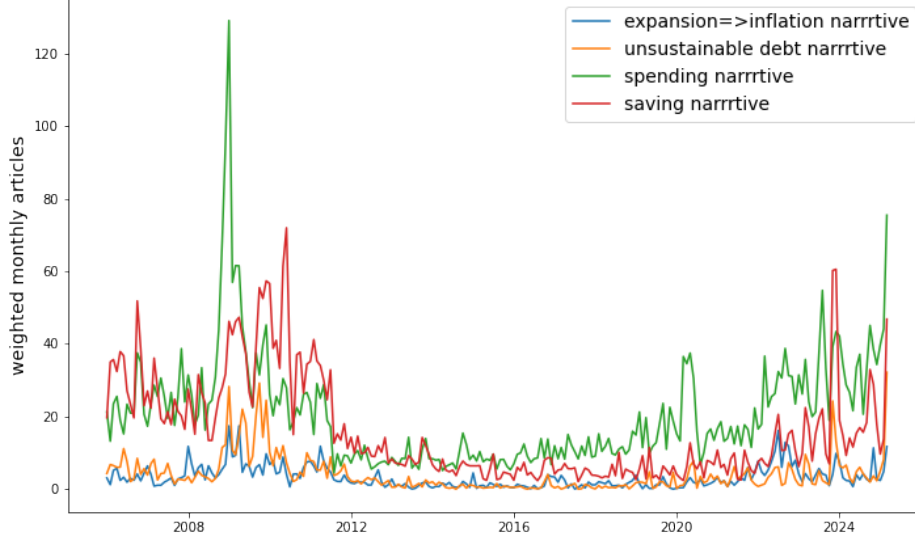


Figure 3: Final narrative indices.

The final narrative indices are the by readership weighted newspaper indices.

Table 2: Descriptive Statistics: Fiscal Narrative Indices

Narrative	Full Sample (2006M1–2025M3)					CES Period (2020M4–2024M12)				
	Mean	SD	Min	Max	N	Mean	SD	Min	Max	N
Expansion → Inflation	3.268	3.104	0.000	17.426	231	4.272	3.311	0.709	16.127	57
Unsustainable Debt	4.004	5.055	0.000	32.180	231	4.025	3.700	0.274	24.286	57
Spending	20.671	15.339	4.492	129.005	231	26.029	10.494	6.761	54.729	57
Saving	15.928	14.254	2.077	71.941	231	13.208	11.047	2.527	60.536	57

Notes: The table reports descriptive statistics for the four readership-weighted fiscal narrative indices. Each observation is one calendar month. The index is constructed as the sum of narrative-classified articles across Bild, FAZ, Süddeutsche Zeitung, and taz, weighted by newspaper readership (IVW 2022). The full sample spans January 2006 to March 2025 ($N = 231$ months). The CES period covers April 2020 to December 2024 ($N = 57$ months), corresponding to the estimation sample. All indices are measured in readership-weighted article counts per month.

Table 2 reports descriptive statistics for the four narrative indices over the full sample and the CES estimation period. The spending and saving narratives are considerably more prevalent than the expansion-inflation and unsustainable debt narratives, reflecting the dominance of fiscal consolidation and public investment debates in German newspaper coverage. The expansion-inflation narrative is more intense during the

CES period than the full sample, consistent with heightened media attention to the inflation consequences of fiscal policy following the 2022 energy crisis.

4 Narrative Index Validation

4.1 Event-based Validity and Editorial Consistency

Figures 4–7 plot the four narrative indices over the full sample period, disaggregated by newspaper and shown together with the readership-weighted aggregate. Two features of these figures support the validity of the classification.

Event-based validity. The indices respond to the fiscal events one would expect them to capture. The unsustainable debt narrative peaks sharply during the eurozone sovereign debt crisis (2009–2010) and rises again around the German Constitutional Court’s debt brake ruling in November 2023, the two episodes in which public debt sustainability was most prominently debated in Germany. The expansion $\Rightarrow$ inflation narrative is largely dormant before 2021 and rises sharply with the post-COVID inflation surge, remaining elevated through 2023. This timing is consistent with the narrative being tied to actual inflationary episodes rather than reflecting general fiscal coverage. The spending narrative exhibits its largest spike around the global financial crisis (2009), when large-scale stimulus packages dominated the policy debate, and increases again after the Russian invasion of Ukraine (2022), when defence and energy expenditure returned to the agenda. The saving narrative displays the opposite pattern: it is elevated throughout the austerity debate of 2006–2011 and declines structurally after the eurozone crisis recedes, consistent with fiscal consolidation falling off the agenda. The 2024 spike in the saving narrative coincides with the collapse of the German coalition government and the subsequent debate over fiscal retrenchment.

Editorial consistency across newspapers. A second source of validation comes from the cross-newspaper heterogeneity visible in the figures. If the classifier is capturing editorial differences rather than noise, newspapers with known ideological positions should differ systematically across narratives. This is what we find. *Süddeutsche Zeitung* and *taz* consistently produce more articles classified under the spending narrative throughout the sample, consistent with their left editorial stance and preference for expansionary fiscal arguments. *Frankfurter Allgemeine Zeitung* contributes disproportionately to the saving narrative, reflecting its ordoliberal orientation and emphasis on fiscal discipline. *Bild* shows lower article counts across all narratives, consistent with its tabloid format and shorter average article length, but contributes substantially to the weighted aggregate due to its dominant readership. These patterns hold across the full sample period and are not driven by specific episodes, suggesting that the classifier is recovering stable editorial differences rather than idiosyncratic classification errors.

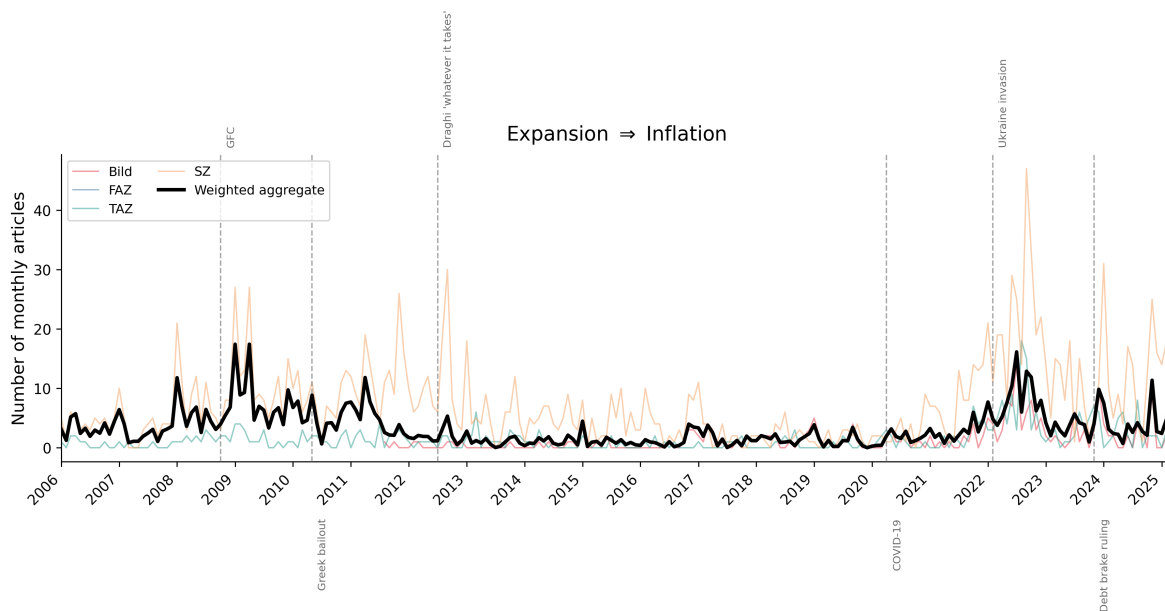


Figure 4: Expansion $\Rightarrow$ Inflation narrative index by newspaper and readership-weighted aggregate, 2006–2025. Dashed vertical lines indicate the global financial crisis (Sep. 2008), Greek bailout (May 2010), Draghi “whatever it takes” speech (Jul. 2012), COVID-19 pandemic (Apr. 2020), Russian invasion of Ukraine (Feb. 2022), and German Constitutional Court debt brake ruling (Nov. 2023).

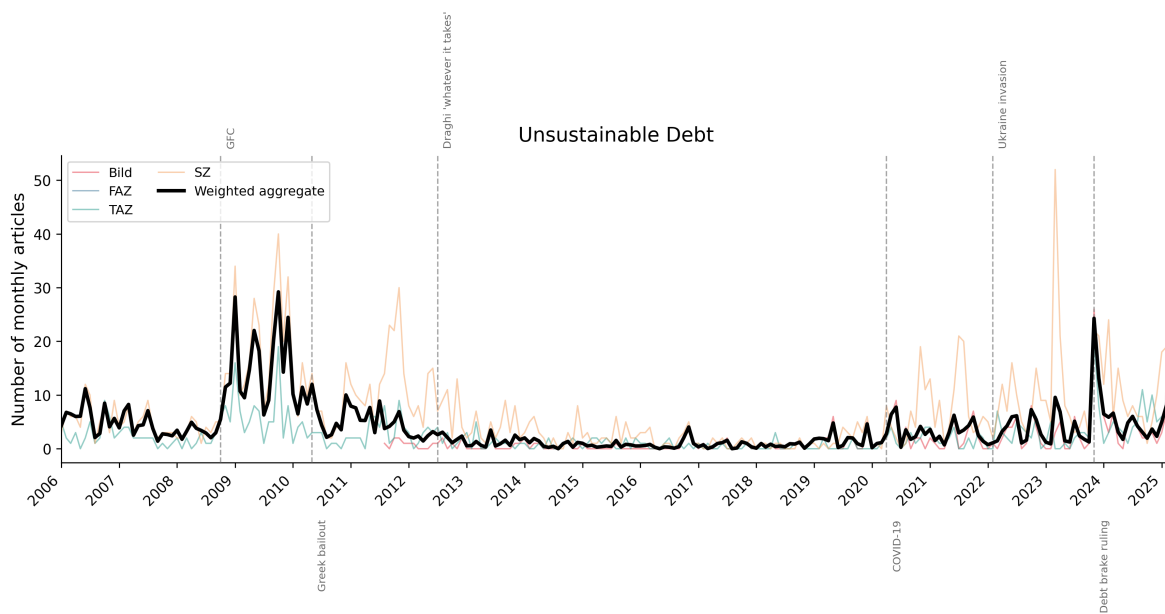


Figure 5: Unsustainable debt narrative index by newspaper and readership-weighted aggregate, 2006–2025. See Figure 4 for event line definitions.

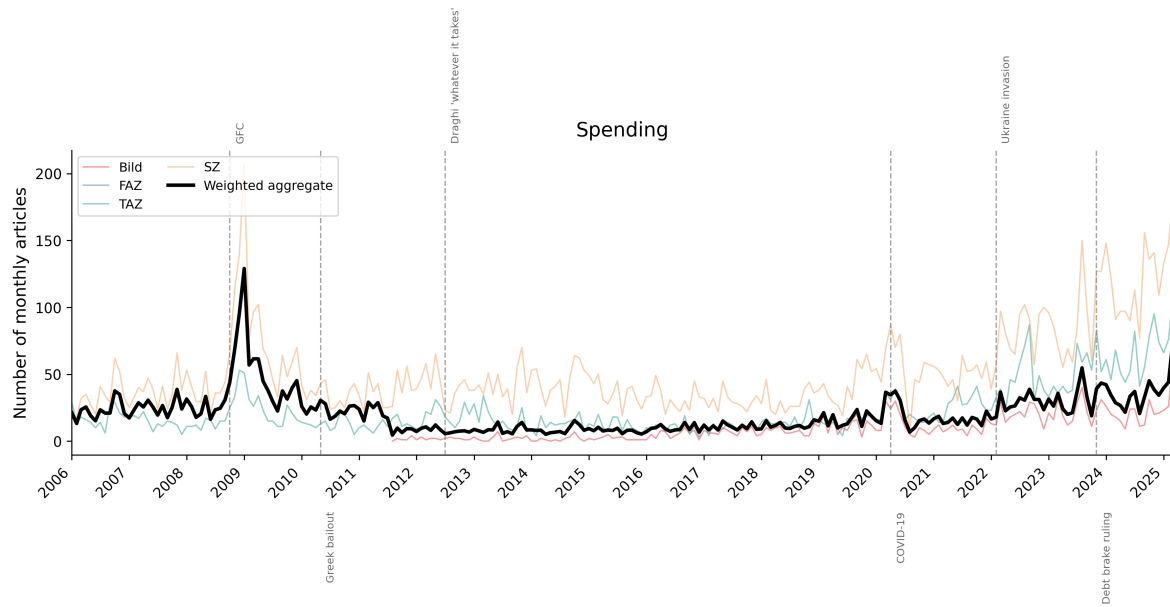


Figure 6: Spending narrative index by newspaper and readership-weighted aggregate, 2006–2025. See Figure 4 for event line definitions.

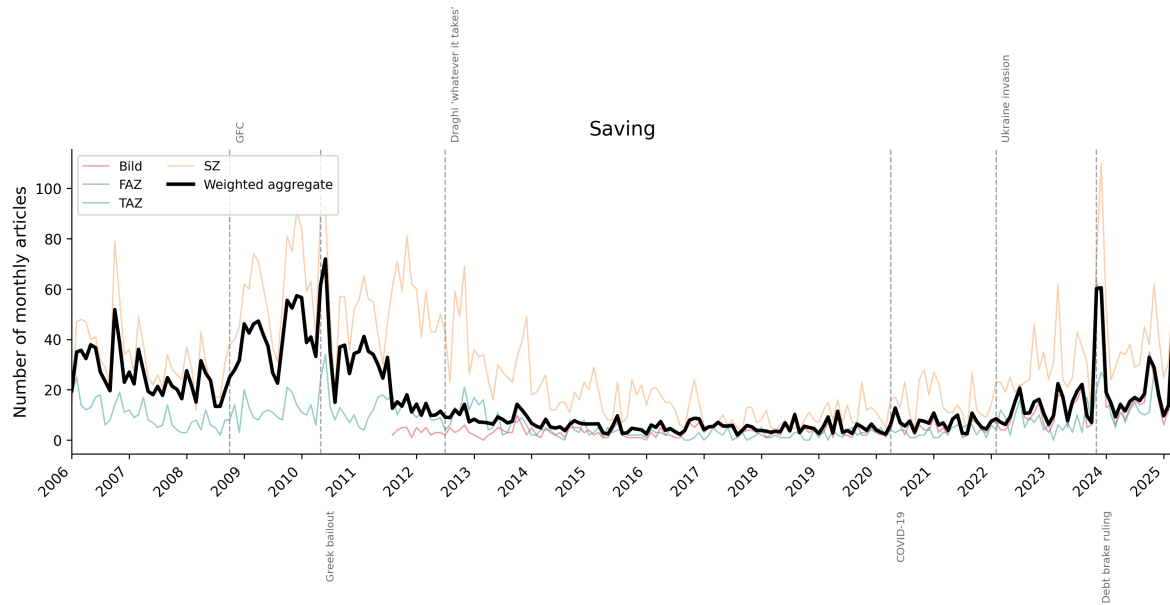


Figure 7: Saving narrative index by newspaper and readership-weighted aggregate, 2006–2025. See Figure 4 for event line definitions.

4.2 Comparison to the Dictionary Approach

To further assess the reliability of the LLM classifier, we compare its output to a simple keyword-based classification applied to the same corpus. For each narrative, we construct a dictionary of German-language terms directly associated with the narrative’s core concept — for example, *Staatsverschuldung*, *Schuldenbremse*, and *Haushaltsdefizit* for the unsustainable debt narrative, and *Sparmaßnahmen*, *Konsolidierung*, and *schwarze Null* for the saving narrative. An article is classified as positive by the keyword classifier if it contains at least one term from the corresponding dictionary. The full keyword lists are reported in Table 4 in the appendix.

Rather than comparing classifiers at the article level, we validate at the level of the monthly indices — the objects that directly enter our regressions. For each newspaper $\times$ narrative combination, we aggregate keyword-positive article counts to the monthly frequency and compute the Pearson and Spearman correlation with the corresponding LLM-based monthly index. Table 3 reports the results.

Across all 16 newspaper $\times$ narrative combinations, Pearson correlations range from 0.39 to 0.85, with an average of 0.68. Spearman rank correlations, which are more robust to outlier months such as the 2009 fiscal stimulus spike, range from 0.36 to 0.83, with an average of 0.63. These figures confirm that the LLM indices are broadly consistent with a transparent, human-readable rule-based benchmark and are not the product of random classification.

The results vary systematically across narratives and newspapers in ways that are themselves informative. The spending and saving narratives achieve the highest correlations for *taz* (0.87 and 0.74) and *Süddeutsche Zeitung* (0.76 and 0.79), consistent with these papers employing relatively direct fiscal vocabulary. *Frankfurter Allgemeine Zeitung* shows notably lower correlations for the spending narrative (Pearson 0.39, Spearman 0.36), while performing well on saving (0.70) and expansion $\Rightarrow$ inflation (0.78). We interpret this pattern as reflecting *FAZ*’s analytical style: its coverage of public expenditure tends to be framed in abstract economic terms rather than through direct expenditure vocabulary, reducing keyword overlap without indicating misclassification. This is precisely the type of case that motivates the use of a language model over a dictionary approach — the LLM captures narratives present in the framing and argument structure of an article rather than in its surface terminology alone.

4.3 Correlation with External Indices

As a third validation exercise, we correlate the readership-weighted aggregate narrative indices with three sets of external indicators: the Ifo Business Climate Index and its components, the fiscal sentiment index of ?, and Google Trends search volume data for narrative-related terms in Germany (forthcoming). Unlike the keyword comparison

Table 3: Correlation Between LLM and Keyword-Based Monthly Indices

Newspaper	Narrative	Pearson r	Spearman ρ	Months
<i>Bild</i>	Expansion $\Rightarrow$ Inflation	0.665	0.598	164
	Unsustainable Debt	0.852	0.464	164
	Spending	0.801	0.833	164
	Saving	0.667	0.596	164
<i>FAZ</i>	Expansion $\Rightarrow$ Inflation	0.778	0.657	230
	Unsustainable Debt	0.476	0.569	230
	Spending	0.387	0.358	230
	Saving	0.703	0.693	230
<i>taz</i>	Expansion $\Rightarrow$ Inflation	0.671	0.556	231
	Unsustainable Debt	0.585	0.551	231
	Spending	0.867	0.713	231
	Saving	0.735	0.742	231
<i>SZ</i>	Expansion $\Rightarrow$ Inflation	0.693	0.709	231
	Unsustainable Debt	0.582	0.652	231
	Spending	0.764	0.667	231
	Saving	0.793	0.825	231
<i>Average</i>	Expansion $\Rightarrow$ Inflation	0.702	0.630	–
	Unsustainable Debt	0.624	0.559	–
	Spending	0.705	0.643	–
	Saving	0.725	0.714	–

Notes: Pearson r and Spearman ρ between the monthly LLM-based article count and the monthly keyword-based article count, by newspaper and narrative. The keyword classifier flags an article as positive if it contains at least one term from the narrative-specific dictionary (see Table 4). Months reports the number of months in the common sample. *Bild* has a shorter sample (164 months) because its article-level data begin later than the other newspapers. Average rows report the unweighted mean across the four newspapers.

Table 4: Keyword Dictionaries by Narrative

Narrative	Keywords
Expansion $\Rightarrow$ Inflation	Inflation, Preissteigerung, Teuerung, Kaufkraft, Konjunkturpaket, Nachfrageinflation, Fiskalimpuls, Überhitzung, Lohn-Preis, Lohninflation, Geldmenge, Fiskalstimulus, Nachfrageüberhang
Unsustainable Debt	Staatsverschuldung, Schuldenbremse, Haushaltsdefizit, Tragfähigkeit, Schuldenstand, Schuldenkrise, Neuverschuldung, Schuldenquote, Staatspleite, Verschuldung, Schuldenproblem, Haushaltslage, Defizit, Insolvenz
Spending	Investitionen, Ausgabenprogramm, Förderprogramm, Subventionen, Haushaltsplan, Mehrausgaben, Infrastrukturausgaben, öffentliche Investitionen, Haushaltsmittel, Ausgabenerhöhung, Ausgabensteigerung, Staatsausgaben, Finanzierung
Saving	Sparen, Konsolidierung, Austerität, Schuldenabbau, Sparmaßnahmen, Sparpaket, Ausgabenkürzung, schwarze Null, Schuldentilgung, Haushaltsdisziplin, Haushaltskonsolidierung, Nullverschuldung, fiskalische Konsolidierung, Haushaltsstabilität

Notes: An article is classified as positive by the keyword classifier if it contains at least one term from the corresponding list. Matching is case-insensitive and requires whole-word matches to avoid spurious partial matches in German compound nouns.

in Section 4.2, which validates the internal consistency of the classifier, these exercises assess whether the narrative indices have macroeconomic content — that is, whether they co-move with independently measured economic conditions and public attention in economically plausible ways.

ifo Business Climate Index. Table 5 reports correlations between the narrative indices and the ifo Business Climate Index and its two components — business situation and business expectations — published monthly by the ifo Institute for Economic Research. All four narrative indices are negatively correlated with the ifo indices, a pattern that is economically interpretable rather than indicative of misclassification. Fiscal narratives — whether about spending, debt, saving, or the inflationary consequences of expansion — tend to intensify during periods of economic stress, precisely when ifo indices decline. This reflects the well-documented tendency for fiscal policy to become salient in public discourse during downturns: large stimulus programmes dominate the spending narrative during recessions, debt sustainability concerns peak during fiscal crises, and inflation narratives are amplified during supply-side shocks that simultaneously depress business sentiment.

The spending narrative exhibits the strongest negative correlation with the ifo Business Climate Index (Pearson $r = -0.68$, Spearman $\rho = -0.55$), consistent with spending coverage peaking during the global financial crisis of 2009 and the COVID-19 pandemic — both periods of sharply declining ifo readings. The unsustainable debt narrative shows the strongest co-movement with the business situation component ($r = -0.57$), reflecting the concentration of debt sustainability concerns during the eurozone sovereign debt crisis of 2009–2012, when current economic conditions deteriorated markedly. By contrast, the saving narrative is essentially uncorrelated with business expectations ($r = -0.06$, $\rho = -0.07$), consistent with fiscal consolidation being discussed in both expansionary periods — when it is treated as prudent policy — and contractionary periods — when it is debated as a response to fiscal stress. The absence of a systematic relationship with forward-looking expectations is therefore economically meaningful rather than a failure of the index.

Latifi et al. (2024) Fiscal Sentiment Index. Latifi et al. (2024) construct a quarterly fiscal sentiment index for Germany using LLM-based classification of parliamentary speeches and newspaper articles, covering the period 2006–2021. Their overall sentiment measure (*senti*) captures the degree of expansionary fiscal discourse, with higher values indicating more expansionary sentiment. We aggregate our monthly narrative indices to quarterly averages and correlate them with their overall sentiment index, its exogenous and endogenous components, and a dictionary-based alternative (*dict1*). Table 6 reports the results.

Table 5: Correlation of Narrative Indices with ifo Business Climate Components

Narrative	Business Climate		Business Situation		Business Expectations	
	r	ρ	r	ρ	r	ρ
Expansion $\Rightarrow$ Inflation	-0.449	-0.383	-0.398	-0.381	-0.400	-0.273
Unsustainable Debt	-0.454	-0.413	-0.569	-0.478	-0.233	-0.179
Spending	-0.682	-0.551	-0.627	-0.554	-0.593	-0.421
Saving	-0.303	-0.339	-0.477	-0.472	-0.055	-0.065

Notes: Pearson r and Spearman ρ between the readership-weighted aggregate narrative index and the ifo Business Climate Index and its two components, published monthly by the ifo Institute for Economic Research. All series at monthly frequency over the common sample 2006–2025 ($N = 231$ months). ifo indices are seasonally adjusted, base year 2015 = 100.

All four narrative indices are negatively correlated with the overall Latifi sentiment measure, consistent with our indices capturing periods of fiscal stress during which expansionary sentiment is low. The saving narrative shows the strongest negative correlation ($r = -0.54$, $\rho = -0.58$), followed by the expansion $\Rightarrow$ inflation narrative ($r = -0.46$, $\rho = -0.49$) and the unsustainable debt narrative ($r = -0.43$, $\rho = -0.44$). The spending narrative shows a weaker overall correlation ($r = -0.24$), but its endogenous component is positively correlated with our spending index ($r = 0.25$, $\rho = 0.39$) — consistent with spending narratives rising endogenously in response to deteriorating economic conditions, which is precisely the mechanism our paper studies.

Particularly informative is the pattern across the endogenous and exogenous components of the Latifi index. The endogenous sentiment component — which captures fiscal discourse driven by macroeconomic feedback — is near zero for all narratives except spending, suggesting that our narrative indices are not mechanically reflecting macro outcomes but are instead capturing shifts in fiscal framing that have an independent informational content. This pattern supports the identification strategy in Section 5, where we exploit variation in narrative exposure across households to isolate the causal effect of fiscal narratives on inflation expectations.

Google Trends. As a final external check, we correlate the narrative indices with Google Trends search volume for narrative-related German-language terms, restricting the sample to searches originating in Germany. Search volume captures public attention to fiscal topics independently of both our classifier and the ifo survey, providing a demand-side validation of the narrative indices. We use monthly search volume for *Inflation* (expansion $\Rightarrow$ inflation), *Staatsverschuldung* (unsustainable debt), *Haushalt* (spending), and *Kürzungen* (saving). Google Trends normalises search volume to a 0–100 scale within each series, where 100 denotes the month of peak search activity; months with very low absolute search volume are suppressed and treated as missing.

Table 6: Correlation of Narrative Indices with Latifi et al. (2024) Fiscal Sentiment Index

Narrative	Overall		Exogenous		Endogenous		Dictionary	
	r	ρ	r	ρ	r	ρ	r	ρ
Expansion $\Rightarrow$ Inflation	-0.457	-0.494	-0.369	-0.414	-0.043	0.004	-0.161	-0.269
Unsustainable Debt	-0.428	-0.435	-0.337	-0.369	0.061	0.129	-0.082	-0.306
Spending	-0.243	-0.238	-0.184	-0.227	0.249	0.392	0.062	0.031
Saving	-0.539	-0.582	-0.429	-0.484	0.033	0.029	-0.408	-0.534

Notes: Pearson r and Spearman ρ between the readership-weighted aggregate narrative index (aggregated to quarterly frequency) and components of the fiscal sentiment index of Latifi et al. (2024). Overall: aggregate fiscal sentiment index. Exogenous: component of sentiment driven by narrative shifts independent of macroeconomic conditions. Endogenous: component driven by macroeconomic feedback. Dictionary: dictionary-based sentiment measure (*dict1*). Common sample: 2006Q1–2021Q3 ($N = 63$ quarters).

Table 7 reports the correlations and Figure 8 plots the two series together for each narrative.

Three of the four narrative indices show meaningful positive correlations with their corresponding search terms. The saving narrative exhibits the strongest co-movement with *Kürzungen* search volume (Pearson $r = 0.52$, Spearman $\rho = 0.65$), consistent with austerity coverage in newspapers generating — or reflecting — heightened public interest in spending cuts, with both series peaking during the eurozone crisis of 2009–2010 and again following the debt brake ruling in late 2023. The expansion $\Rightarrow$ inflation narrative co-moves closely with *Inflation* searches ($r = 0.48$, $\rho = 0.52$): both series are dormant before 2021 and spike sharply during the post-pandemic inflation surge of 2022–2023. The unsustainable debt narrative shows moderate correlation with *Staatsverschuldung* searches ($r = 0.43$, $\rho = 0.35$), with co-movement most visible during the eurozone sovereign debt crisis. The spending narrative shows a weaker correlation with *Haushalt* ($r = 0.18$, $\rho = 0.16$), which is unsurprising given that *Haushalt* is a generic budget term that peaks annually during the parliamentary budget debate in autumn, irrespective of whether spending narratives are elevated. Across all four narratives, the positive direction of the correlations confirms that the indices capture fiscal discourse that resonates with and is reflected in public information-seeking behaviour.

Fiscal Narrative Indices and Google Trends Search Volume

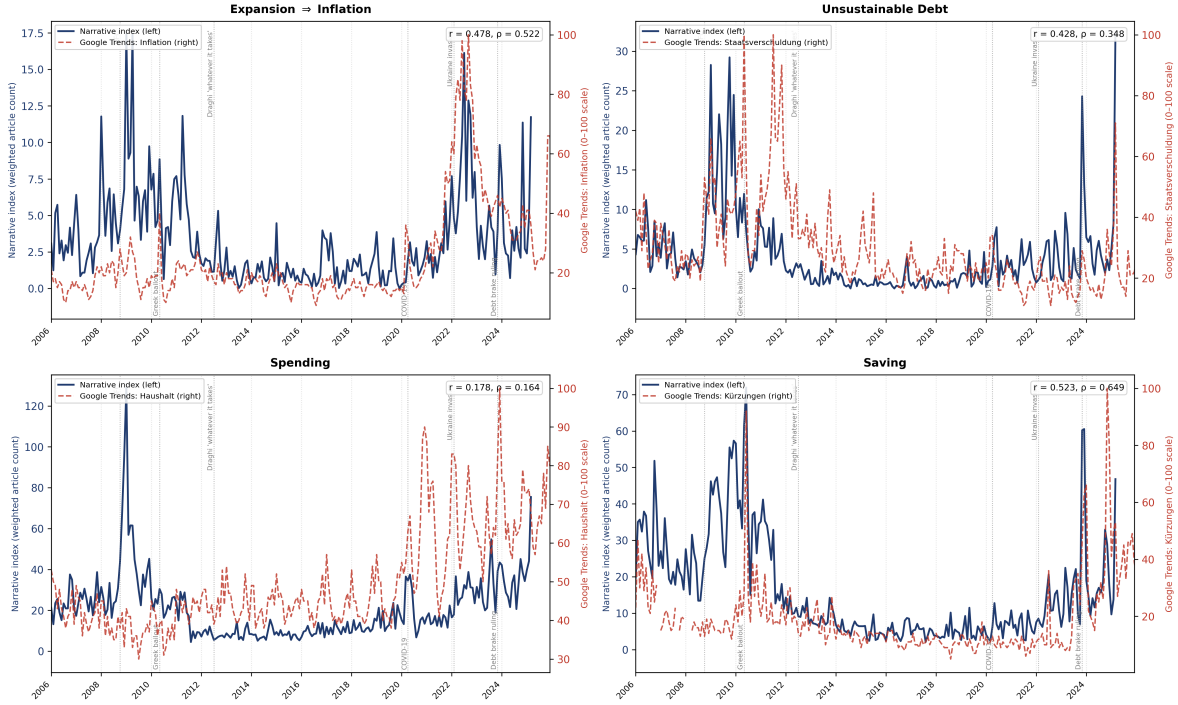


Figure 8: Fiscal Narrative Indices and Google Trends Search Volume

Notes: Each panel plots the readership-weighted aggregate narrative index (blue solid line, left axis) against monthly Google Trends search volume for the corresponding search term (red dashed line, right axis), restricted to searches originating in Germany. Google Trends values are normalised to a 0–100 scale within each series; note that the right axis scale varies across panels, reflecting differences in the minimum search volume floor across terms. Vertical dotted lines mark six key events: the global financial crisis (Oct. 2008), the Greek bailout (May 2010), Draghi’s “whatever it takes” speech (Jul. 2012), the onset of the COVID-19 pandemic (Apr. 2020), the Russian invasion of Ukraine (Feb. 2022), and the German Constitutional Court’s debt brake ruling (Nov. 2023). Pearson r and Spearman ρ are reported in the top right corner of each panel.

Table 7: Correlation of Narrative Indices with Google Trends Search Volume

Narrative	Search Term	Pearson r	Spearman ρ	N
Expansion $\Rightarrow$ Inflation	<i>Inflation</i>	0.478	0.522	231
Unsustainable Debt	<i>Staatsverschuldung</i>	0.428	0.348	231
Spending	<i>Haushalt</i>	0.178	0.164	231
Saving	<i>Kürzungen</i>	0.523	0.649	227

Notes: Pearson r and Spearman ρ between the readership-weighted aggregate narrative index and monthly Google Trends search volume for the listed term, restricted to searches originating in Germany. Google Trends normalises search volume to a 0–100 scale within each series; months with very low absolute search volume are suppressed and excluded from the correlation. Common sample: 2006M1–2025M3. N denotes the number of months with non-missing observations for both series.

5 Fiscal narratives’ effect on inflation expectations

One mechanism of how fiscal narratives in the media potentially influence prices and other macro variables is by affecting household’s inflation expectations. While it is quite common in theoretical models for inflation expectations to have an effect on actual inflation (see e.g., [Clarida et al., 1999](#); [Smets, 2003](#); [Woodford & Walsh, 2005](#)), [Moessner \(2025\)](#) also provides empirical evidence for it. In the following we are testing the hypothesis, whether the fiscal narratives we identify in the media increase household inflation expectations and therefor potentially affect actual inflation.

5.1 Data and model specification

For the estimation we use the European Central Bank’s (ECB) Consumer Expectations Survey (CES), which runs since April 2020. On a monthly basis, the CES asks between 2,000 and 3,000 participants in Germany for their quantitative inflation expectations: *“How much higher (lower) do you think prices in general will be 12 months from now in the country you currently live in? Please give your best guess of the change in percentage terms. You can provide a number up to one decimal place.”* The survey micro data is available online and provides background information on participants’ age, gender, and education and income level. Our sample spans from April 2020 to December 2024.

To measure the effect of the prevalence of the four fiscal narratives in the media on households’ 12-months ahead inflation expectations denoted by $E_{it}\pi_{t+12}$, we exploit the panel dimension of the CES and estimate the following model:

$$E_{it}\pi_{t+12} = c + \alpha_i + \beta Narrative_{t-1}^j + \gamma X_{t-1} + \epsilon_{it}. \quad (1)$$

Individual households are indexed by $i = 1, \dots, n$ and the time dimension by $t = 1, \dots, T$. We include household fixed effects α_i to control for time-invariant household characteristics, including baseline inflation pessimism, habitual newspaper readership, income level, and risk attitudes. Following [Bańkowska et al. \(2021\)](#), we winsorize the sample at the 2% and 98% level per survey wave to prevent extreme responses from driving the results. The vector X_t contains a set of time-varying aggregate macro controls: the German year-on-year inflation rate (π_t) to capture the prevailing inflation environment, industrial production (IP_t) as a monthly proxy for the business cycle, the ECB’s policy rate (ECB_rate_t) to absorb the monetary policy stance, the Ifo Business Climate Index (ifo_klima_t) to control for forward-looking business sentiment that might jointly drive editorial decisions and household beliefs, the German unemployment rate ($unemp_rate_t$) to account for labour market conditions, which are a well-established determinant of inflation expectations (?), the HICP energy price index ($energy_idx_t$) — particularly relevant given the 2021–2022 energy price spike which

simultaneously triggered fiscal policy debates in the media and directly raised household inflation expectations — the 10-year Bund yield ($bund_yield_t$) to absorb sovereign risk perceptions that may independently drive both debt sustainability narratives and household beliefs about future price levels, a Cholesky-identified government spending shock ($fiscal_shock_t$) derived from a structural VAR estimated on quarterly German macro data to control for actual fiscal impulses that may generate narrative coverage as a byproduct, and the total number of fiscal policy related articles weighted by readership ($total_fiscal_t$) to ensure that effects are not driven by an overall increase in media attention to fiscal policy. The variable $Narrative_t^j$ denotes one of the four fiscal narrative indicators j described above and shown for the survey sample period in Figure 9. In Appendix D we also report results for the individual newspapers. Finally, ϵ_{it} is the error term.

The narrative indicators and control variables both enter the regression model with a one-period lag, because some news may have been published after the household’s survey participation and most macro data is released with a lag. The data collection for each wave of the CES typically starts on the first Thursday of a month and closes on the first Tuesday of the following month. For example, in the September 2024 wave, households were surveyed between September 5 and 30. Eurostat published a first flash estimate of August inflation on August 30, and the final August inflation data on September 18, 2024. All variables enter the estimation in levels; only industrial production is transformed to log differences. We assume no reverse causality: households are influenced by the media, but no individual household has an impact on the number of articles sharing a specific fiscal policy narrative.

Identification. A key concern in estimating Equation 1 is that both narrative coverage and household inflation expectations may respond to common macroeconomic shocks, potentially inflating the estimated $\hat{\beta}$. We address this through three complementary strategies.

Household fixed effects. The inclusion of α_i absorbs all time-invariant household-level confounders, including baseline inflation pessimism, habitual newspaper readership, income level, risk attitudes, and financial literacy. Identification comes entirely from within-household variation over time: we ask whether a given household’s inflation expectations are higher in months when fiscal narrative coverage is elevated, relative to that household’s own average expectations. This rules out any story in which households with persistently high expectations also happen to live in high-narrative environments.

Total fiscal coverage. The variable $total_fiscal_{t-1}$ captures the overall volume of fiscal policy articles across all newspapers, weighted by readership. Including it as a control means that $\hat{\beta}$ is identified from variation in the *composition* of fiscal narratives rather than from the overall level of media attention to fiscal policy. A month with ele-

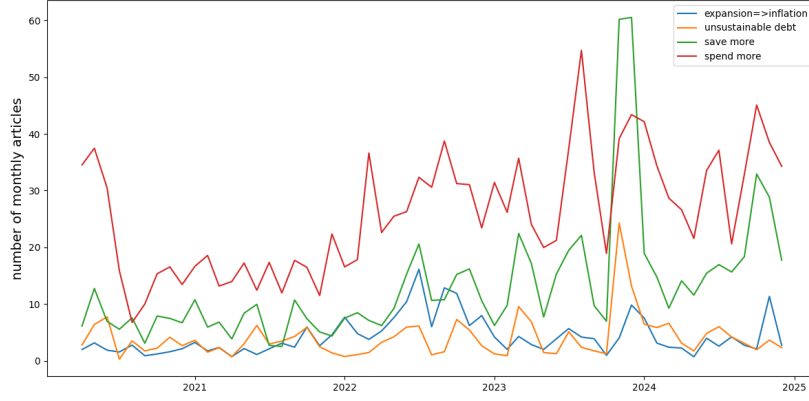


Figure 9: This figure displays the four aggregate fiscal narrative indicators from April 2020 to December 2024. Each series represents the weighted number of monthly articles across FAZ, Bild, SZ, and TAZ falling into the i) fiscal expansion leads to inflation (blue), ii) public debt is unsustainably high (orange), iii) the government should increase savings (green), or iv) its spending (red) narrative.

vated spending coverage but low saving coverage differs from a month with the reverse pattern even when total fiscal salience is identical. This conditional variation in narrative composition is substantially less likely to be driven by aggregate macroeconomic stress than the overall level of fiscal coverage itself.

Month-of-year fixed effects. In a standard panel regression, time fixed effects would absorb all aggregate shocks in each period. However, our narrative indices vary only at the monthly aggregate level, so including full time fixed effects would be collinear with the treatment by construction. Instead, we include month-of-year fixed effects, which absorb systematic seasonal variation in inflation expectations — for example, households consistently report higher expectations in spring and autumn regardless of narrative content. The remaining identification assumption is that, conditional on household fixed effects, the full set of time-varying macro controls, and seasonal dummies, the narrative composition index is uncorrelated with unobserved aggregate shocks to inflation expectations. The stability of $\hat{\beta}$ as controls are added sequentially (Table ??, columns 1–5) provides empirical support for this assumption.

5.2 Results

Table 8 reports the main results of Model 1 for the four fiscal narrative indicators. Each column presents a separate fixed-effects regression including the full set of macro controls and month-of-year fixed effects. Full regression output for each narrative is reported individually in Appendix C, Tables C-1 to C-4. Clustered standard errors at the household level are reported in parentheses.

For three out of four narrative indicators we find statistically significant positive effects on 12-month ahead household inflation expectations. As reported in column (1)

Table 8: Effect of Fiscal Narratives on 12-Month Ahead Inflation Expectations

	(1) Exp→Infl	(2) Debt Sus.	(3) Save	(4) Spend
$Narrative_{t-1}$	0.055*** (0.009)	0.004 (0.005)	0.011*** (0.002)	0.018*** (0.003)
π_{t-1}	-0.072*** (0.017)	-0.057*** (0.017)	-0.047*** (0.017)	-0.060*** (0.017)
IP_{t-1}	-0.512 (0.366)	-0.317 (0.369)	-0.269 (0.365)	-0.860** (0.373)
ECB_{rate}_{t-1}	-0.704*** (0.055)	-0.803*** (0.053)	-0.795*** (0.053)	-0.821*** (0.053)
$total_fiscal_{t-1}$	0.001** (0.001)	0.003*** (0.001)	0.001 (0.001)	0.001* (0.001)
ifo_klima_{t-1}	-0.070*** (0.005)	-0.080*** (0.005)	-0.080*** (0.005)	-0.061*** (0.006)
$fiscal_shock_{t-1}$	0.289*** (0.037)	0.366*** (0.038)	0.337*** (0.038)	0.429*** (0.040)
$unemp_rate_{t-1}$	-0.862*** (0.097)	-0.954*** (0.098)	-0.915*** (0.097)	-0.770*** (0.100)
$energy_idx_{t-1}$	0.045*** (0.005)	0.043*** (0.005)	0.044*** (0.004)	0.045*** (0.004)
$bund_yield_{t-1}$	0.230** (0.100)	0.327*** (0.104)	0.313*** (0.098)	0.336*** (0.098)
Household FE	Yes	Yes	Yes	Yes
Month-of-year FE	Yes	Yes	Yes	Yes
Observations	142,761	142,761	142,761	142,761

Notes: The dependent variable is 12-month ahead inflation expectations winsorized at the 2nd and 98th percentile within each survey wave. Each column reports a separate fixed-effects regression for one of the four fiscal narrative indices. “ $Narrative_{t-1}$ ” refers to the respective readership-weighted narrative index: expansion→inflation in column (1), unsustainable debt in column (2), saving in column (3), and spending in column (4). All variables enter with a one-period lag. Standard errors clustered at the household level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

of Table 8, an additional readership-weighted article linking fiscal expansion to inflation increases household inflation expectations by 0.055 pp. The standard deviation of the expansion→inflation narrative over the CES sample period is 3.31, implying that a one standard deviation increase in narrative intensity raises inflation expectations by approximately 0.18 pp. The effects of the macro controls are in line with economic theory: higher ifo business climate reduces expectations, higher energy prices raise them, and tighter monetary policy (higher ECB rate) lowers them.

The effect of articles deeming government debt to be unsustainable (column 2) is positive but statistically insignificant at conventional levels ($\hat{\beta} = 0.004$, $p = 0.484$). This result should be interpreted alongside the subsample analysis in Section ??, which shows that the debt sustainability narrative becomes strongly significant in the post-2022 period ($\hat{\beta} = 0.148$, $p < 0.001$), suggesting a state-dependent channel that is activated only when inflation is already salient.

Media coverage of fiscal consolidation and spending also has significantly positive effects on household expectations. Column (3) reports an increase of 0.011 pp per additional saving narrative article, and column (4) reports an increase of 0.018 pp per additional spending narrative article. With standard deviations of 11.0 and 10.5 respectively over the CES period, one standard deviation increases translate into 0.12 pp and 0.19 pp rises in inflation expectations.

The baseline results are obtained using winsorized survey data. We also estimate Model 1 on non-adjusted data; results reported in Appendix E show that the estimates are not materially sensitive to winsorization. For further robustness we repeat the estimation without weighting newspaper data by readership; results are reported in Appendix F. Only the effect of the unsustainable debt narrative becomes insignificant under equal weighting, consistent with Bild’s high readership weight driving the identification for that narrative.

5.2.1 Joint Effects of Fiscal Narratives

Table 9 reports the joint effects of all four narrative indicators on households’ 12-month ahead inflation expectations, estimated simultaneously in a single regression including the full set of macro controls and month-of-year fixed effects.

Three out of four narratives remain statistically significant when included jointly. The expansion leads to inflation narrative retains a positive and highly significant effect ($\hat{\beta} = 0.059$), slightly larger than in the individual specification, suggesting that controlling for the other narratives sharpens identification of this channel. The saving and spending narratives also remain positive and significant, with magnitudes close to the individual estimates.

The unsustainable debt narrative, however, turns negative and significant ($\hat{\beta} = -0.022$, $p = 0.005$) once the other three narratives are included simultaneously. This

sign reversal reflects the high correlation between the debt sustainability narrative and the other narratives.

Overall, the joint specification confirms that expansionary and inflation-related fiscal narratives are the most consistently associated with higher household inflation expectations, even after controlling for the competing narrative frames and a rich set of macroeconomic variables.

5.3 Robustness Checks

5.3.1 Wave-Level and Two-Way Clustering

A potential concern with our baseline specification is that standard errors clustered at the household level may understate the true uncertainty in our estimates. Since the narrative index $Narrative_{t-1}^j$ varies only at the monthly aggregate level, all households surveyed in the same wave are exposed to identical narrative values. This within-wave correlation in the treatment variable means that household-clustered standard errors may be overly optimistic, treating observations within the same wave as more independent than they actually are.

To address this, we re-estimate the baseline specification using two alternative clustering schemes. First, we cluster at the survey wave level, which treats all observations within the same monthly wave as potentially correlated and accounts directly for the aggregate nature of the treatment variable. With 52 survey waves in our sample, this is the most conservative standard error adjustment available. Second, we implement two-way clustering at both the household and survey wave level simultaneously, following [Cameron et al. \(2011\)](#). This accounts for both within-household serial correlation over time and within-wave cross-sectional correlation due to the common narrative exposure.

Table 10 reports the results alongside the baseline household-clustered estimates for comparison. The coefficients are identical across all three clustering schemes by construction — only the standard errors change. Wave-level clustering substantially widens the standard errors, as expected given that the treatment varies only at the wave level and there are only 52 clusters. Under wave-level clustering, the expansion→inflation narrative remains significant at the 10% level ($\hat{\beta} = 0.055$, $SE = 0.031$), as does the saving narrative ($\hat{\beta} = 0.011$, $SE = 0.006$). The spending narrative loses conventional significance ($\hat{\beta} = 0.018$, $SE = 0.012$, $p = 0.129$), while the unsustainable debt narrative remains insignificant as in the baseline. Two-way clustering produces nearly identical results to wave-level clustering, confirming that the dominant source of uncertainty is within-wave correlation rather than within-household serial correlation.

The reduction in significance under wave-level clustering reflects the fundamental identification challenge of aggregate treatment variables in household panel settings, and is not specific to our paper — it arises whenever a time-varying aggregate variable is

Table 9: Joint Effects of Fiscal Narratives on $E_{it}\pi_{t+12}$

	(1)
<i>expansion_inflation</i> _{<i>t</i>-1}	0.059*** (0.010)
<i>debt_unsustainable</i> _{<i>t</i>-1}	-0.022*** (0.008)
<i>saving</i> _{<i>t</i>-1}	0.013*** (0.003)
<i>spending</i> _{<i>t</i>-1}	0.017*** (0.004)
π_{t-1}	-0.074*** (0.018)
<i>IP</i> _{<i>t</i>-1}	-0.876** (0.397)
<i>ECB_rate</i> _{<i>t</i>-1}	-0.735*** (0.056)
<i>total_fiscal</i> _{<i>t</i>-1}	-0.002*** (0.001)
<i>ifo_klima</i> _{<i>t</i>-1}	-0.049*** (0.006)
<i>fiscal_shock</i> _{<i>t</i>-1}	0.304*** (0.044)
<i>unemp_rate</i> _{<i>t</i>-1}	-0.664*** (0.102)
<i>energy_idx</i> _{<i>t</i>-1}	0.047*** (0.005)
<i>bund_yield</i> _{<i>t</i>-1}	0.294*** (0.108)
Household FE	Yes
Month-of-year FE	Yes
Observations	142,761
Groups	13,997
Within R^2	0.047

Notes: The dependent variable is 12-month ahead inflation expectations winsorized at the 2nd and 98th percentile within each survey wave. All four fiscal narrative indices enter simultaneously. All variables enter with a one-period lag. Standard errors clustered at the household level in parentheses. The sign reversal of the debt sustainability narrative relative to the individual specification reflects multicollinearity with the expansion→inflation narrative; the individual estimate in Table 8 is the preferred specification for that channel. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

used as treatment in a micro panel (see [Moulton, 1990](#)). Importantly, the direction and magnitude of all four narrative effects are fully robust to the choice of clustering scheme, and the expansion→inflation and saving narratives remain statistically significant even under the most conservative specification.

Table 10: Robustness: Alternative Clustering Schemes

	(1) HH cluster	(2) Wave cluster	(3) Two-way cluster
<i>Panel A: Expansion → Inflation</i>			
Narrative _{t-1}	0.055*** (0.009)	0.055* (0.031)	0.055* (0.030)
<i>Panel B: Unsustainable Debt</i>			
Narrative _{t-1}	0.004 (0.005)	0.004 (0.013)	0.004 (0.012)
<i>Panel C: Saving</i>			
Narrative _{t-1}	0.011*** (0.002)	0.011* (0.006)	0.011* (0.006)
<i>Panel D: Spending</i>			
Narrative _{t-1}	0.018*** (0.003)	0.018 (0.012)	0.018 (0.011)
Household FE	Yes	Yes	Yes
Month-of-year FE	Yes	Yes	Yes
Macro controls	Yes	Yes	Yes
Cluster	HH	Wave	HH × Wave
Observations	138,965	138,600	138,600

Notes: Each panel reports the narrative coefficient from a separate fixed-effects regression with the full set of macro controls and month-of-year fixed effects. Column (1) replicates the baseline specification with standard errors clustered at the household level. Column (2) clusters at the survey wave level (52 clusters), accounting for the aggregate nature of the treatment variable. Column (3) uses two-way clustering at the household and wave level simultaneously following [Cameron et al. \(2011\)](#). Coefficients are identical across columns by construction. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

5.3.2 Double Machine Learning

As a further robustness check, we estimate the effect of fiscal narratives on inflation expectations using Double Machine Learning (DML) following ?. DML addresses two potential concerns with our baseline specification: first, that the control function may be misspecified if the relationship between macro controls and expectations is nonlinear; second, that the choice of which controls to include may itself introduce bias. By using machine learning to flexibly partial out controls from both the outcome and the treatment, DML provides a specification-robust estimate of the narrative effect.

Implementation. We implement a partial linear model in which the treatment D_{it} is the demeaned narrative index and the outcome Y_{it} is demeaned household inflation expectations. Demeaning is performed at the household level prior to estimation, absorbing household fixed effects in the same way as the within estimator in our baseline specification. We specify the partial linear model as:

$$Y_{it} = \theta D_{it} + g(X_{t-1}) + \epsilon_{it} \quad (2)$$

where $g(\cdot)$ is an unknown function of the macro controls X_{t-1} estimated via machine learning. We use 5-fold cross-fitting with 3 resamples to obtain stable estimates, competing OLS and rigorous Lasso (?) as candidate learners for both the outcome equation $E[Y|X]$ and the treatment equation $E[D|X]$. The final DML estimate $\hat{\theta}$ is obtained by regressing the outcome residual on the treatment residual from the best-performing learner selected by cross-validated mean squared error.

Learner selection. In all four narrative specifications, the cross-validation procedure selects OLS over Lasso for both the outcome and treatment equations across all three resamples. This finding is itself informative: it implies that the relationship between macro controls and both inflation expectations and narrative indices is well-approximated by a linear model. Lasso does not improve out-of-sample prediction over OLS, confirming that our baseline linear specification is not suffering from functional form misspecification. This result is consistent with the PDS-Lasso robustness check in Section ??, where nonlinear terms provided only modest additional explanatory power.

Selected controls. The standalone Lasso runs — which we use to inspect which controls are informative for each equation — reveal that the outcome equation selects seven out of eight controls consistently across all narratives, dropping only the 10-year Bund yield. The treatment equations show more heterogeneity: the debt sustainability narrative requires all eight controls to predict, reflecting its complex dependence on the full macro environment; the saving narrative is most strongly predicted by unemployment and inflation dynamics; the spending narrative drops the ECB rate; and the expansion→inflation narrative drops the energy price index. This heterogeneity in the treatment equations confirms that each narrative is driven by a distinct combination of macroeconomic conditions, providing further support for treating the four narratives as conceptually distinct channels.

Results. Table 11 reports the DML estimates for all four narratives. The coefficients are 0.064 (expansion→inflation), 0.019 (unsustainable debt), 0.015 (saving), and 0.013 (spending), all statistically significant at the 1% level. These estimates are close to the

baseline FE results in Table 8, with the small differences attributable to the larger sample available for DML — which does not require the fiscal shock variable and therefore retains 155,130 observations compared to 142,761 in the main specification. The stability of the narrative coefficients across the FE and DML estimators, combined with the cross-validation evidence that OLS dominates Lasso, provides strong evidence that the baseline results are not driven by omitted nonlinear confounding or control selection bias.

Table 11: Double Machine Learning Estimates

	(1) Exp→Infl	(2) Debt Sus.	(3) Save	(4) Spend
Narrative _{t-1}	0.064*** (0.006)	0.019*** (0.004)	0.015*** (0.002)	0.013*** (0.002)
Learner Y	OLS	OLS	OLS	OLS
Learner D	OLS	OLS	OLS	OLS
Cross-fit folds	5	5	5	5
Resamples	3	3	3	3
Household FE	Yes	Yes	Yes	Yes
Observations	155,130	155,130	155,130	155,130

Notes: Double Machine Learning estimates following ?. Household fixed effects absorbed via within-household demeaning prior to estimation. Candidate learners are OLS and rigorous Lasso; the reported learner is the one selected by minimum cross-validated MSE. Macro controls include demeaned inflation, industrial production, ECB policy rate, ifo business climate index, unemployment rate, HICP energy price index, 10-year Bund yield, and total fiscal articles. Standard errors are robust. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

5.3.3 State-Dependence: Pre- and Post-2022 Subsamples

A key prediction of rational inattention models is that households pay more attention to inflation-relevant information when inflation is already salient (??). To test this, we split the sample at February 2022 — the onset of the energy and inflation crisis in Germany — and re-estimate the baseline specification separately for the pre-crisis period (April 2020 to January 2022, waves 4–33) and the post-crisis period (February 2022 onwards, waves 34–60). Table 12 reports the results.

The findings reveal striking state-dependence across all four narratives. The expansion→inflation narrative exhibits the most dramatic shift: the coefficient is negative and significant in the pre-crisis period ($\hat{\beta} = -0.051$, $p < 0.001$) and large, positive, and highly significant post-crisis ($\hat{\beta} = 0.322$, $p < 0.001$). The sign reversal in the pre-2022 period is consistent with households interpreting fiscal expansion narratives as stabilising during the COVID period — when fiscal stimulus was broadly welcomed as crisis relief — rather than as inflationary. Once inflation became salient post-2022, the same narrative con-

tent triggered upward expectation revisions, as households connected fiscal expansion to the inflationary pressures they were already experiencing.

The debt sustainability narrative is insignificant pre-2022 ($\hat{\beta} = 0.023$, $p = 0.140$) but strongly significant post-2022 ($\hat{\beta} = 0.148$, $p < 0.001$). This confirms the rational inattention interpretation: concerns about public debt sustainability are too abstract to influence expectations when inflation is low, but become highly relevant when households are already alert to price pressures and fiscal sustainability concerns are directly linked to inflationary financing risks.

The saving narrative shows a different pattern — it is significant and of similar magnitude in both periods ($\hat{\beta} = 0.039^{***}$ pre-2022 and $\hat{\beta} = 0.039^{***}$ post-2022). This stability suggests that saving narratives operate through a channel that is active regardless of the inflation regime, possibly reflecting a general fiscal stress signal that households consistently interpret as price-relevant.

The spending narrative is marginally significant pre-2022 ($\hat{\beta} = 0.014$, $p = 0.078$) and strongly significant post-2022 ($\hat{\beta} = 0.176$, $p < 0.001$), with the coefficient increasing more than tenfold. This amplification is consistent with the transmission results in Section ??: spending narratives prime households to expect higher demand and prices, and this priming effect is substantially stronger when inflation is already elevated.

Taken together, the subsample results support a narrative state-dependence hypothesis: fiscal narratives have larger and more consistent effects on inflation expectations when the inflation environment makes such narratives immediately relevant. This is particularly pronounced for the expansion→inflation, debt sustainability, and spending narratives, which are all insignificant or smaller in the low-inflation pre-2022 period and large and significant post-2022.

Table 12: State-Dependence: Pre- and Post-2022 Subsample Split

	Exp→Infl		Debt Sus.		Save		Spend	
	Pre (1)	Post (2)	Pre (3)	Post (4)	Pre (5)	Post (6)	Pre (7)	Post (8)
Narrative _{t-1}	-0.051*** (0.013)	0.322*** (0.051)	0.023 (0.016)	0.148*** (0.024)	0.039*** (0.009)	0.039*** (0.006)	0.014* (0.008)	0.176*** (0.028)
Household FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Month-of-year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Macro controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	72,667	66,298	72,667	66,298	72,667	66,298	72,667	66,298

Notes: Each column reports the narrative coefficient from a separate fixed-effects regression with the full set of macro controls and month-of-year fixed effects. The pre-crisis sample covers April 2020 to January 2022 (survey rounds 4–33). The post-crisis sample covers February 2022 onwards (survey rounds 34–60). Standard errors clustered at the household level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

5.3.4 Excluding Major Fiscal Announcement Months

A potential concern is that our narrative indices spike around major fiscal policy announcements, and that the estimated effects on inflation expectations may be driven by these specific high-salience episodes rather than by the general narrative channel. To address this, we exclude observations from survey waves that coincide with major fiscal announcements in Germany during our sample period: the first COVID stimulus package (wave 6, June 2020), the coalition agreement (wave 31, July 2022), the Doppelwumms energy relief package (wave 35, November 2022), the Bundeswehr special fund announcement (wave 41, May 2023), the Schuldenbremse constitutional court ruling (wave 55, July 2024), and the coalition collapse (wave 58, October 2024). This removes 16,502 observations, leaving a sample of 124,463.

Table 13 reports the results alongside the baseline estimates. The narrative coefficients are virtually unchanged across all four specifications. The expansion→inflation coefficient falls marginally from 0.055*** to 0.046*** but remains highly significant. The saving narrative falls slightly from 0.011*** to 0.009*** and remains significant. The spending narrative is fully stable at 0.018*** versus 0.018*** in the baseline. The unsustainable debt narrative remains insignificant in both specifications (0.004 versus 0.004).

These results confirm that the narrative effects are not driven by a small number of high-profile announcement episodes but instead reflect the persistent influence of the broader fiscal narrative environment on household inflation expectations throughout the sample period.

5.4 Heterogeneity Analysis

5.4.1 Gender

In Table 14, we extend the baseline specification by interacting each fiscal narrative with the respondent’s gender. Our gender-interaction estimates show that women react significantly more strongly to several fiscal narratives.

For the inflation narrative, the baseline effect is positive and significant, and the interaction term is sizable and strongly significant. This means that when newspapers emphasize that expansion leads to inflation, women revise their inflation expectations upward more sharply than men.

For spending narratives, the results show a similar asymmetry. The main effect of spending-related news is positive and highly significant, but the interaction term indicates that women react even more strongly to news advocating for the government to spend more.

By contrast, for the saving and debt-sustainability narratives, the interaction terms

Table 13: Robustness: Excluding Major Fiscal Announcement Months

	(1) Exp→Infl	(2) Debt Sus.	(3) Save	(4) Spend
<i>Panel A: Baseline</i>				
Narrative _{t-1}	0.055*** (0.009)	0.004 (0.005)	0.011*** (0.002)	0.018*** (0.003)
Observations	138,965	138,965	138,965	138,965
<i>Panel B: Excluding announcement months</i>				
Narrative _{t-1}	0.046*** (0.009)	0.004 (0.006)	0.009*** (0.002)	0.018*** (0.003)
Observations	124,463	124,463	124,463	124,463
Household FE	Yes	Yes	Yes	Yes
Month-of-year FE	Yes	Yes	Yes	Yes
Macro controls	Yes	Yes	Yes	Yes

Notes: Panel A replicates the baseline specification. Panel B excludes observations from survey waves coinciding with major fiscal policy announcements: the first COVID stimulus package (wave 6), the coalition agreement (wave 31), the Doppelwumms energy relief package (wave 35), the Bundeswehr special fund (wave 41), the Schuldenbremse constitutional court ruling (wave 55), and the coalition collapse (wave 58). Standard errors clustered at the household level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

are small and statistically insignificant, implying no detectable gender differences in how these fiscal themes influence inflation expectations.

Finally, this pattern is consistent with the literature documenting gender differences in inflation expectations and in sensitivity to economic information (D’Acunto et al. (2020), Chalwadi et al. (2023), Corduas (2022)).

5.4.2 Age

Table 15 examines whether the relationship between fiscal narratives and inflation expectations varies systematically across age groups. The results reveal a clear and economically meaningful pattern for inflation-related narratives, whereas heterogeneity is largely absent for the remaining fiscal narratives.

First, inflation narratives exhibit substantial age heterogeneity. While the baseline coefficient remains positive and statistically significant, the interaction terms for respondents aged 35–49 and 50–70 are likewise positive and highly significant. This indicates that middle-aged individuals adjust their inflation expectations more strongly when exposed to narratives emphasizing inflationary fiscal conditions. By contrast, the oldest group (71+) shows no significant additional responsiveness. These findings suggest that narrative-driven inflation signals are most salient for individuals who are typically more economically active—those with mortgages, financial commitments, or wage-setting concerns—and therefore more attentive to inflation-relevant information.

Table 14: Gender Heterogeneity in the Effect of Fiscal Narratives on Inflation Expectations

	(1) Inflation	(2) Debt Sustainability	(3) Saving	(4) Spending
Narrative _{t-1}	0.082*** (0.008)	0.008* (0.005)	0.013*** (0.002)	0.029*** (0.002)
Female × Narrative _{t-1}	0.050*** (0.013)	0.003 (0.007)	-0.001 (0.003)	0.012*** (0.004)
π_{t-1}	0.211*** (0.010)	0.261*** (0.010)	0.269*** (0.010)	0.246*** (0.010)
IP_{t-1}	-3.657*** (0.287)	-3.265*** (0.286)	-3.111*** (0.285)	-2.340*** (0.282)
ECB_rate_{t-1}	-0.077*** (0.024)	-0.205*** (0.023)	-0.194*** (0.023)	-0.195*** (0.023)
Fiscal tone _{t-1}	0.000 (0.000)	0.004*** (0.000)	0.002*** (0.000)	0.000 (0.000)
Household FE	yes	yes	yes	yes
Observations	159,783	159,783	159,783	159,783
R ² (within)	0.033	0.031	0.031	0.033

Notes: Standard errors clustered at the household level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Narratives are aggregated newspaper mentions weighted by sentiment. Gender is a female dummy (1 = female).

Second, debt sustainability narratives do not display meaningful age-related heterogeneity. Neither the direct effect nor the interaction terms are statistically significant across age categories. This implies that concerns about the long-run fiscal position of the government do not differentially shape inflation expectations across the life cycle.

Third, saving-related narratives likewise show no heterogeneity by age. Despite a statistically significant baseline effect, the interaction terms across all age groups remain small and insignificant, suggesting a uniform response to messages about fiscal prudence.

Finally, spending narratives reveal only weak and statistically insignificant differences across age groups. Although the baseline coefficient is positive and robust—consistent with expansionary narratives raising households' inflation expectations—the additional effect by age group is negligible.

Taken together, the evidence suggests that narratives linking fiscal policy to inflation outcomes uniquely and disproportionately influence the expectations of middle-aged individuals, while other fiscal narratives affect inflation expectations in a broadly homogeneous way across the age distribution. This provides further support for the view that exposure, salience, and economic engagement shape how individuals internalize fiscal news when forming inflation beliefs.

Table 15: Age heterogeneity in the effect of fiscal narratives on $E_{it}\pi_{t+12}$

	(1) Inflation	(2) Debt sustainability	(3) Saving	(4) Spending
Narrative $_{t-1}$	0.051*** (0.014)	0.013 (0.008)	0.013*** (0.004)	0.029*** (0.004)
Age 35–49 $\times$ Narrative $_{t-1}$	0.070*** (0.018)	-0.003 (0.010)	-0.002 (0.004)	0.009 (0.005)
Age 50–70 $\times$ Narrative $_{t-1}$	0.076*** (0.018)	-0.007 (0.010)	0.000 (0.004)	0.007 (0.005)
Age 71+ $\times$ Narrative $_{t-1}$	0.021 (0.024)	0.002 (0.013)	0.002 (0.005)	-0.004 (0.007)
π_{t-1}	0.212*** (0.010)	0.260*** (0.010)	0.268*** (0.010)	0.246*** (0.010)
IP_{t-1}	-3.653*** (0.287)	-3.266*** (0.286)	-3.109*** (0.285)	-2.330*** (0.282)
ECB_rate_{t-1}	-0.076*** (0.024)	-0.206*** (0.023)	-0.194*** (0.023)	-0.195*** (0.023)
Fiscal tone $_{t-1}$	0.000 (0.000)	0.004*** (0.000)	0.002*** (0.000)	0.000 (0.000)
Household FE	yes	yes	yes	yes
Observations	159,783	159,783	159,783	159,783
R ² (within)	0.033	0.031	0.031	0.033

Notes: Standard errors clustered at the household level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. The reference age group is 18–34 (category 1). “Narrative” refers to the corresponding aggregate fiscal narrative index in each column (inflation, debt sustainability, saving, and spending).

5.5 Discussion

The relationship between household inflation expectations, and the fiscal expansion leads to inflation narrative is in line with what one would expect. The more articles report on fiscal expansion leading to inflation, the more households will adapt this narrative for themselves and also expect higher inflation. Given that the point estimate with only household fixed effects falls substantially when adding controls (from 0.203 to 0.105, Table ??), this suggests that much of the effect is channeled through general economic sentiment and other macro-shocks. Once controlling for the standard drivers of expectations, the effect appears to be robust but economically modest. The standard deviation of the fiscal expansion $\Rightarrow$ inflation indicator on the CES sample length is 3.31 articles. This means that a standard deviation uptick in articles following this narratives translates into an increase of inflation expectations by around 0.35 pp. As is usual in micro-level FE regressions, the overall R^2 is low—household expectations are driven by many idiosyncratic factors. Yet the narrative effect remains highly statistically

significant once we condition on standard macro drivers.

The positive coefficient of the unsustainable government debt narrative indicator in column (5) of Table ?? is also in line with what [Bassetto & Miller \(2025\)](#) predict. An increase in coverage deeming public debt to be unsustainably high leads to more households acquiring this information which they conclude can result in sudden inflation scares. This effect is, though statistically significant, in absolute terms much smaller than for the fiscal expansion $\Rightarrow$ inflation narrative. The standard deviation in the survey period is 3.7 such that an increase of articles of this size translates into a increase in inflation expectations by 0.03 pp. The saving narrative is closest to the unsustainable debt narrative and its coefficient being 0.013 is just slightly bigger.

6 The effect of fiscal narratives on the transmission government spending shocks

Motivated by the result that fiscal narratives can increase household inflation expectations, we next analyze their effect on the transmission of government spending shocks to real GDP and inflation. We do so by following a two-step estimation approach. In the first step, we estimate a structural government spending shock and in the second step, we employ the resulting shock series in local projections. We model non-linearity of the transmission of a government spending shock by interacting it with the narrative indicators, one at a time.

6.1 Estimation strategy

We estimate the government spending shock series in a structural VAR (SVAR) model employing a recursive identification strategy with short-run restrictions. In the selection of variables and ordering we follow [Auerbach & Gorodnichenko \(2012\)](#) by placing a measure of expected government spending first, to control for fiscal foresight ($FG_{t|t-1}$), followed by government spending (G_t), tax revenues (τ_t), and GDP (Y_t). $FG_{t|t-1}$ is ordered first because a contemporaneous shock in G_t cannot affect a forecast for period t made in $t-1$ (see [Auerbach & Gorodnichenko \(2012\)](#) and [Leduc et al. \(2007\)](#) for further discussion). The ordering of the rest of the variables follows a standard specification first introduced by [Blanchard & Perotti \(2002\)](#). Using the Akaike Information Criterion we determine the optimal number of lags to be 8.

In the data selection for the vector of variables $X_t = [FG_{t|t-1}, \ln(G_t), \ln(\tau_t), \ln(Y_t)]'$ in the VAR, we follow [Amendola et al. \(2020\)](#). Government spending is the sum of the seasonally adjusted series of real general government final consumption expenditure and real gross fixed capital formation for Germany, tax revenues are provided by the German Federal Statistical Office as total revenues across the federal and local levels. As

a measure of fiscal foresight, we use the OECD government spending growth forecasts as proposed by [Ilori et al. \(2022\)](#). The OECD forecasts are produced twice a year and are published in the publicly available *OECD Economic Outlook*. We use the government spending forecast data provided by [Ilori et al. \(2022\)](#), which ends in 2019Q4 and extend it forward, interpolating the semi-annual data to quarterly frequency by applying mid-point averaging. All series but the government spending forecasts enter the model in logs and the frequency is quarterly.

In the second step, we employ the government spending shock estimated in the VAR in a local projections model as in [Jordà \(2005\)](#). To model the potentially non-linear effect the fiscal narrative has on the transmission of the government spending shock, we introduce an interaction term ($shock_t \cdot narrative_t$):

$$y_{t+h} = \alpha_h + \beta_h shock_t + \gamma_h(shock_t \cdot narrative_t) + \sum_{l=1}^n \delta_{h,l} W_{t-l} + u_{t+h} \quad (3)$$

The interaction term is in the spirit of [Ascari & Haber \(2022\)](#), who interact a monetary policy shock variable with its absolute values to model the non-linear effect of the shock size on its transmission. Analogously to [Ascari & Haber \(2022\)](#), the coefficient β_h captures the baseline effect of the government spending shock ($shock_t$), irrespective of the prevalent fiscal narrative, while γ_h measures how this effect changes in the event of a unit-change in the narrative indicator ($narrative_t$). As we standardize the narrative indicators before interacting them with the shock series, this unit-change corresponds to one standard deviation. If γ_h has the opposite sign of β_h , the fiscal narrative dampens the effect of government spending on the variable of interest y_t , and if both have the same sign, the effect will be amplified in the respective direction.

We estimate model 3 for horizons $h = 1, \dots, 12$ which corresponds to three years due to the quarterly data frequency. The control variables W_t comprise up to $n = 4$ lags (motivated by the quarterly frequency of the data) of real GDP, government spending and tax revenues (transformed to logs and multiplied by 100), government spending forecasts, the ECB's shadow rate, constructed by [Wu & Xia \(2020\)](#), the GDP deflator, as well as the shock series itself. This choice follows [Montiel Olea et al. \(2025\)](#) who suggest to include lags of the impulse and control variable as well as other variables that are strong predictors of both in the set of controls to counteract potential dynamic misspecifications as well as imperfections in the shock measure. The reported error bands are computed with Newey-West standard errors to control for autocorrelation.

6.2 Baseline Results

6.2.1 Structural VAR

Figure 10 plots the Cholesky-identified structural shock to government spending. The series exhibits sizable quarter-to-quarter variation, with several large negative shocks around 2009, 2013, and 2020, consistent with known periods of fiscal consolidation or crisis-driven adjustments.

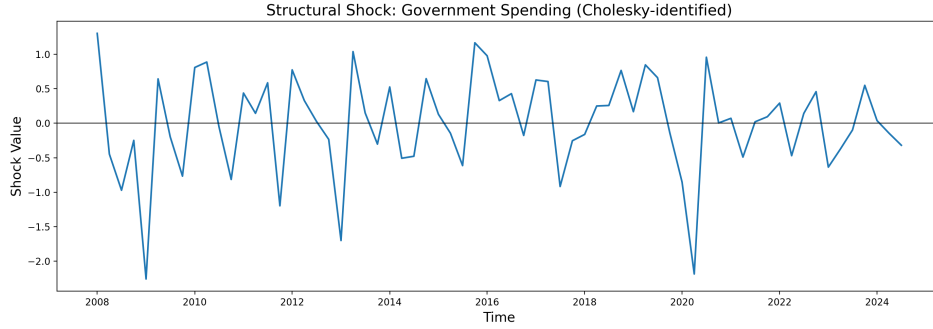


Figure 10: This figure plots the identified structural government spending shock derived from the recursively ordered SVAR.

6.2.2 The effect of individual narrative indicators

Figures 11 and 12 present the results of the local projections described above, measuring the effect of a government spending shock on GDP and inflation, respectively. The blue responses are the linear effects of the fiscal shock while the orange ones depict the interaction term between the shock and one of the four narrative indicators. One can see that each of them act differently on the transmission of a fiscal shock.

The narrative, that the government is supposed to decrease its spending (the saving narrative) has a dampening effect on impact and towards the end of the projection horizon on the transmission of a fiscal shock to GDP. Similarly it has such a dampening effect on inflation in the first four quarters after impact, as the linear and the interaction term coefficients have opposite signs. The opposite is the case for the narrative that the government should increase its spending, for which both have the same sign and hence this narrative amplifies the transmission of a fiscal shock to both GDP and inflation.

The effect of the unsustainable debt narrative on the fiscal shock transmission to GDP seems to be negligible, as the interaction term coefficient rarely differs from zero. In those cases it does, it has the same sign as the linear coefficient, which means that if there is a non-linear effect of this narrative, then it seems to be amplifying the effect of a fiscal shock. This effect is even more pronounced for inflation. The narrative, that fiscal expansion leads to inflation has very limited effects on the transmission of a fiscal shock, as its interaction term coefficient rarely significantly differs from zero.

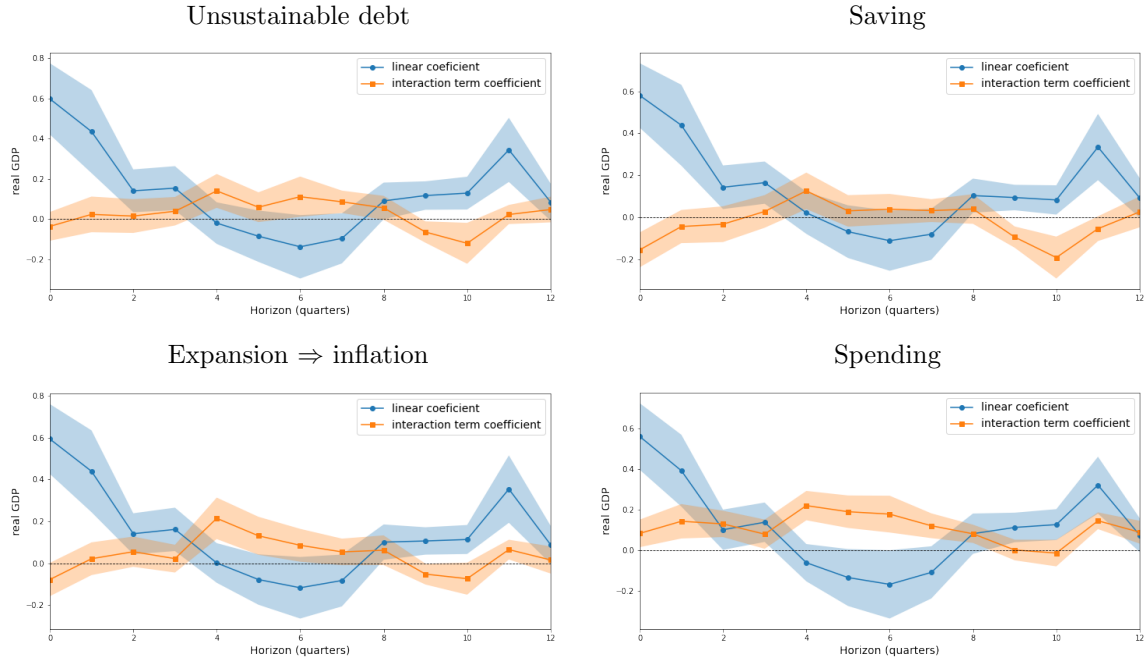


Figure 11: This figure shows the impulse responses of real GDP after and government spending shock (blue) and its interaction (orange) with one of the four narrative indicators. Responses are estimated across a 12 quarters projection horizon and are depicted with 68% confidence bands computed with Newey-West standard errors.

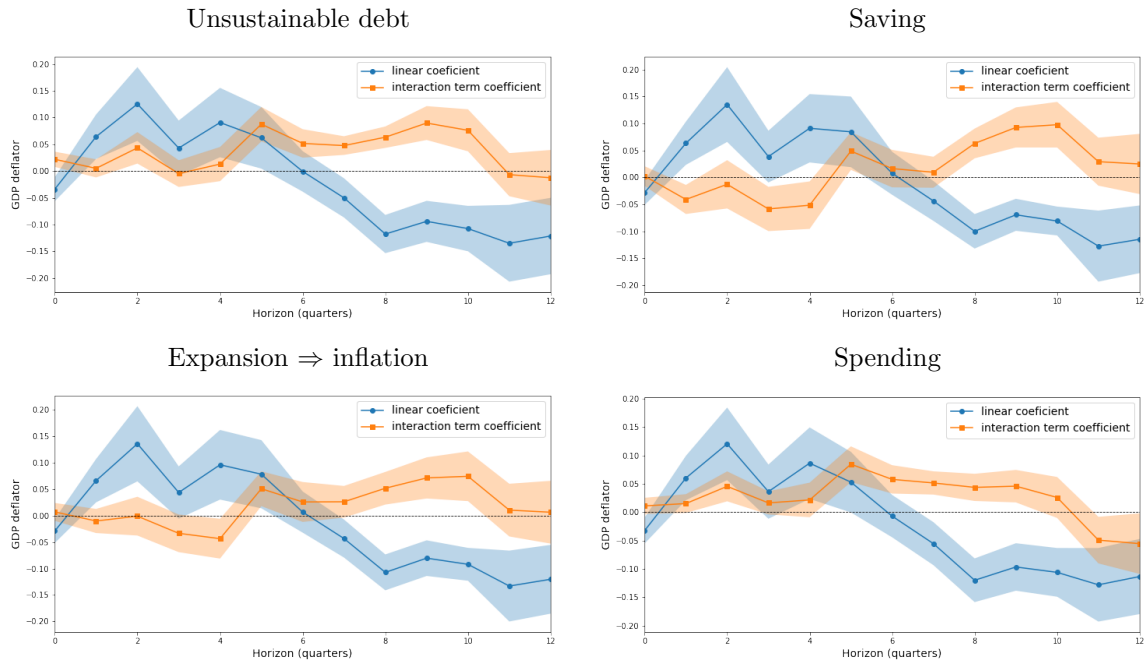


Figure 12: This figure shows the impulse responses of the GDP deflator after and government spending shock (blue) and its interaction (orange) with one of the four narrative indicators. Responses are estimated across a 12 quarters projection horizon and are depicted with 68% confidence bands computed with Newey-West standard errors.

6.2.3 The effect of a composite narrative indicator

Based on the results of the effect of the individual narrative indicators on the transmission of government spending shocks, we create a composite index that signals if government spending shocks will be amplified by which fiscal narratives news outlets convey. We create the composite index using the two narratives that show the clearest and most consistent transmission effects: the spending narrative, which unambiguously amplifies the effect of a fiscal shock on both GDP and inflation, and the saving narrative, which unambiguously dampens it. We therefore define the composite index as the sum of the spending narrative and the saving narrative multiplied by -1 , and standardize the resulting series. This parsimonious construction ensures that the composite captures the narrative dimension most directly linked to fiscal shock transmission. As a robustness check, we also report results for a composite index that additionally includes the unsustainable debt and expansion→inflation narratives in Appendix G. These results look very similar.

We present the resulting index in Figure 13. One can see that the index had very low values in the aftermath of the GFC and at the beginning of the European sovereign debt crisis in 2010. Then, since 2012, the index is constantly increasing with a local peak during COVID-19 and heightened volatility after Russia invaded Ukraine, which sparked a large debate on increasing military spending in Germany. Figure 14 shows that the resulting composite indicator indeed amplifies the effect of a government spending shock on both GDP and inflation. This means, that if the index takes higher values, then the effects of government spending will be larger. Hence, in such periods, a government spending shock is likely to be more inflationary and at the same time more stimulating for output.

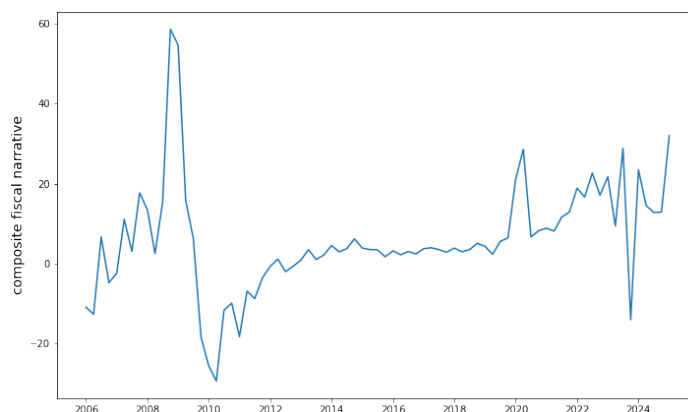


Figure 13: Composite fiscal narrative index. The index is constructed as the sum of the spending narrative (weighted positively) and the saving narrative (weighted negatively), standardized to unit variance.

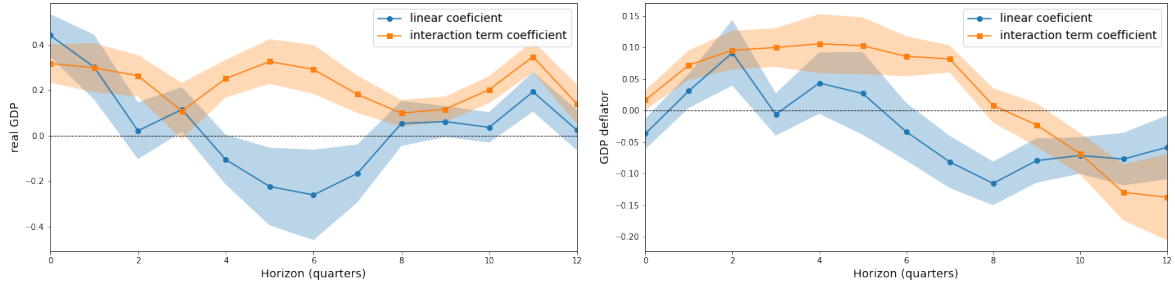


Figure 14: This figure presents the impulse responses of a government spending shock (blue) on GDP (left panel) and the GDP deflator (right panel) and its interaction with the composite fiscal narrative indicator (orange) together with 68% confidence intervals.

7 Conclusion

In this paper, we analyze a large set of articles from German newspapers using AI, and evaluate the fiscal narratives conveyed in these articles. Our results show that these fiscal narratives are not merely descriptive features of public discourse, but play an active role in influencing household inflation expectations. In particular, we find that articles linking fiscal expansion to inflation have the strongest effect, raising household inflation expectations by 0.055 percentage points. Narratives emphasizing government spending or fiscal restraint also have statistically significant, though smaller, impacts of 0.018 and 0.011 percentage points respectively. The effect of debt sustainability narratives is state-dependent, becoming strongly significant in the post-2022 high-inflation regime ($\hat{\beta} = 0.148^{***}$), consistent with rational inattention. These findings highlight the importance of the information environment and the expectations channel in the transmission of fiscal policy to the real economy.

Going further, we demonstrate that the prevalence of specific fiscal narratives in the media can either amplify or dampen the macroeconomic impact of government spending shocks. The spending narrative amplifies the effects of fiscal shocks on GDP and inflation, while the saving narrative dampens them. By combining these two indicators into a composite fiscal narrative index — spending weighted positively and saving weighted negatively — we construct a real-time measure that can signal periods when fiscal shocks are likely to have stronger effects on output and prices. A robustness version of the composite that additionally includes the unsustainable debt and expansion→inflation narratives yields very similar results and is reported in Appendix G.

Overall, our findings underline the critical role of public narratives in shaping economic expectations and policy effectiveness. The composite index we develop offers policymakers a new tool for monitoring the fiscal information climate and anticipating the potential impact of fiscal interventions. More broadly, our approach demonstrates how textual analysis and AI can be harnessed to better understand the evolving interplay between public discourse, expectations, and the macroeconomy.

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Appendix

A Fiscal Dictionaries

General fiscal terms: Fiskal*, Haushalt*, Finanz*, Finanzminist*, Staatsfinanzierung, Staatsbudget, Haushaltsplan

Fiscal expansion: Staatsausgabe, Staatsverschuldung, Kreditaufnahme, Schuldenpolitik, Staatsdefizit, Ausgabenerhöhung, Investition, Subvention, Konjunkturprogramm, Konjunkturpolitik, Wachstumsimpuls, Konjunkturförderung, Finanzspritze, Sondervermögen, Steuersenkung, Prämie

Fiscal consolidation: Haushaltskonsolidierung, Konsolidier*, Schuldenabbau, Sparpolitik, Sparmaßnahme, Defizitreduzierung, Haushaltsdisziplin, Budgetkürzung, Haushaltsüberschuss, Steuererhöhung, Ausgabenkürzung, Ausgabensenkung, Austerität*, Schuldenbremse, Budgetkonsolidierung, Kostensenkung, Sparhaushalt, Rentenkürzung

B ChatGPT prompts

You will be provided with a news article. Your task is to determine whether the article establishes a narrative that fiscal expansion (e.g., government spending increases, tax cuts) will lead to inflation. Classify an article as Yes if it clearly presents a narrative—whether stated as a fact or as a speculation—that fiscal expansion will cause or eventually result in inflation. Do not flag articles where fiscal expansion and inflation are mentioned independently without linking them in a cause-and-effect manner. Do not hypothesize a relationship based on economic theory or indirect reasoning.

Steps:

1. Read the article carefully.
2. Look for any statement or narrative—explicit, combined, or speculative—where fiscal expansion is connected to inflation as a cause or expected outcome.
3. If the article only discusses these topics in isolation or does not establish a cause-and-effect link, classify it as No.
4. Respond with: Response: Yes or No, Explanation (brief, 1-2 sentences): Justify your answer by quoting or summarizing the relevant text that establishes (or does not establish) the connection.

Examples: 1. Article Excerpt: The government's recent increase in spending is expected to drive up prices; experts warn that this fiscal expansion will likely result in higher inflation rates.

Response: Yes,

Explanation: The article clearly links increased government spending to higher inflation through an explicit narrative.

2. Article Excerpt: While government spending has increased, some economists argue that other factors, like global supply chain issues, are keeping inflation in check.

Response: No,

Explanation: The article mentions fiscal expansion and inflation separately but does not establish a cause-and-effect link.

3. Article Excerpt: If the government continues to boost spending, it might eventually set off a chain reaction that results in inflation.

Response: Yes,

Explanation: The article speculates that continued fiscal expansion could eventually lead to inflation, thus establishing the desired narrative.

Article:

Figure B-1: Prompt to assess if an article establishes a causal link between fiscal expansion and inflation.

Analyze the following article to determine whether it suggests that government debt levels in Germany are unsustainable. Consider both explicit statements and implicit impressions, while also accounting for speculative or contradictory arguments. Classify the article into one of the following three categories:

'Yes': If the article states or strongly implies that German government debt levels are unsustainable now or in the near future.

'No': If the article mentions government debt but does not suggest that it is unsustainable, OR if it only raises speculative concerns without claiming current unsustainability.

'NA': If the article does not mention government debt at all, OR if it is not directly mentioning government debt in Germany. If the article discusses economic risks (e.g., inflation, recession) without directly linking them to government debt, classify as 'NA' unless the connection is explicitly made.

Respond with only one of the three categories: 'Yes', 'No', or 'NA'. Provide no additional text or explanation.

Article:

Figure B-2: Prompt to assess if an article conveys that the debt level is unsustainable.

Analyze the following article to determine whether it explicitly conveys an opinion on whether the German government should save more or increase spending. Classify the article into one of the following three categories:

'Save': If the article explicitly states or strongly argues that the German government should reduce spending, save more, or focus on fiscal restraint.

'Spend': If the article explicitly states or strongly argues that the German government should spend more, invest in public programs, or prioritize economic stimulus.

'NA': If the article does not express an explicit opinion on German government saving or spending. The opinion must be explicitly stated - general discussions or reporting on government spending without a clear stance should be classified as 'NA'. If the article presents both perspectives, classify based on the dominant opinion expressed.

Respond with only one of the three categories: 'Save', 'Spend', or 'NA'. Provide no additional text or explanation.

Article:

Figure B-3: Prompt to analyze whether an article expresses the opinion that the government should increase its savings or spending.

C Individual Narratives Regression Tables

Table C-1: Effect of Expansion $\rightarrow$ Inflation Narrative on $E_{it}\pi_{t+12}$: Sequential Controls

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
<i>expansion_inflation</i> _{<i>t</i>-1}	0.203*** (0.006)	0.107*** (0.005)	0.111*** (0.005)	0.105*** (0.005)	0.105*** (0.006)	0.064*** (0.006)	0.055*** (0.009)
π_{t-1}		0.205*** (0.009)	0.203*** (0.009)	0.212*** (0.010)	0.212*** (0.010)	-0.041** (0.016)	-0.072*** (0.017)
<i>IP</i> _{<i>t</i>-1}			-3.652*** (0.286)	-3.663*** (0.286)	-3.663*** (0.287)	-2.146*** (0.286)	-0.512 (0.366)
<i>ECB_rate</i> _{<i>t</i>-1}				-0.078*** (0.023)	-0.077*** (0.024)	-0.568*** (0.040)	-0.704*** (0.055)
<i>total_fiscal</i> _{<i>t</i>-1}					-0.000 (0.000)	-0.001*** (0.000)	0.001** (0.001)
<i>ifo_klima</i> _{<i>t</i>-1}						-0.045*** (0.004)	-0.070*** (0.005)
<i>unemp_rate</i> _{<i>t</i>-1}						-0.531*** (0.066)	-0.862*** (0.097)
<i>energy_idx</i> _{<i>t</i>-1}						0.058*** (0.004)	0.045*** (0.005)
<i>bund_yield</i> _{<i>t</i>-1}						-0.100 (0.076)	0.230** (0.100)
<i>fiscal_shock</i> _{<i>t</i>-1}							0.289*** (0.037)
Household FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Month-of-year FE	No	No	No	No	No	No	Yes
Observations	159,783	159,783	159,783	159,783	159,783	159,783	142,761

Notes: Standard errors clustered at the household level in parentheses. Column (7) adds the Cholesky-identified government spending shock and month-of-year fixed effects; the reduction in observations reflects the sample period covered by the structural VAR. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table C-2: Effect of Unsustainable Debt Narrative on $E_{it}\pi_{t+12}$: Sequential Controls

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
<i>debt_unsustainable</i> _{<i>t</i>-1}	0.013*** (0.003)	0.022*** (0.003)	0.024*** (0.003)	0.032*** (0.003)	0.009** (0.004)	0.019*** (0.004)	0.004 (0.005)
π_{t-1}		0.273*** (0.009)	0.273*** (0.009)	0.283*** (0.010)	0.261*** (0.010)	-0.016 (0.016)	-0.057*** (0.017)
<i>IP</i> _{<i>t</i>-1}			-3.131*** (0.285)	-3.274*** (0.286)	-3.265*** (0.286)	-1.876*** (0.287)	-0.317 (0.369)
<i>ECB_rate</i> _{<i>t</i>-1}				-0.150*** (0.023)	-0.205*** (0.023)	-0.682*** (0.039)	-0.803*** (0.053)
<i>total_fiscal</i> _{<i>t</i>-1}					0.004*** (0.000)	-0.000 (0.000)	0.003*** (0.001)
<i>ifo_klima</i> _{<i>t</i>-1}						-0.053*** (0.004)	-0.080*** (0.005)
<i>unemp_rate</i> _{<i>t</i>-1}						-0.522*** (0.068)	-0.954*** (0.098)
<i>energy_idx</i> _{<i>t</i>-1}						0.058*** (0.004)	0.043*** (0.005)
<i>bund_yield</i> _{<i>t</i>-1}						-0.037 (0.076)	0.327*** (0.104)
<i>fiscal_shock</i> _{<i>t</i>-1}							0.366*** (0.038)
Household FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Month-of-year FE	No	No	No	No	No	No	Yes
Observations	159,783	159,783	159,783	159,783	159,783	159,783	142,761

Notes: Standard errors clustered at the household level in parentheses. Column (7) adds the Cholesky-identified government spending shock and month-of-year fixed effects; the reduction in observations reflects the sample period covered by the structural VAR. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table C-3: Effect of Saving Narrative on $E_{it}\pi_{t+12}$: Sequential Controls

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
<i>saving</i> _{<i>t</i>-1}	0.019*** (0.001)	0.014*** (0.001)	0.014*** (0.001)	0.019*** (0.001)	0.013*** (0.002)	0.015*** (0.002)	0.011*** (0.002)
π_{t-1}		0.269*** (0.009)	0.269*** (0.009)	0.279*** (0.010)	0.269*** (0.010)	-0.005 (0.016)	-0.047*** (0.017)
<i>IP</i> _{<i>t</i>-1}			-2.936*** (0.284)	-3.030*** (0.284)	-3.111*** (0.285)	-1.670*** (0.285)	-0.269 (0.365)
<i>ECB_rate</i> _{<i>t</i>-1}				-0.173*** (0.023)	-0.194*** (0.023)	-0.673*** (0.040)	-0.795*** (0.053)
<i>total_fiscal</i> _{<i>t</i>-1}					0.002*** (0.000)	-0.002*** (0.000)	0.001 (0.001)
<i>ifo_klima</i> _{<i>t</i>-1}						-0.054*** (0.004)	-0.080*** (0.005)
<i>unemp_rate</i> _{<i>t</i>-1}						-0.504*** (0.067)	-0.915*** (0.097)
<i>energy_idx</i> _{<i>t</i>-1}						0.057*** (0.004)	0.044*** (0.004)
<i>bund_yield</i> _{<i>t</i>-1}						-0.015 (0.076)	0.313*** (0.098)
<i>fiscal_shock</i> _{<i>t</i>-1}							0.337*** (0.038)
Household FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Month-of-year FE	No	No	No	No	No	No	Yes
Observations	159,783	159,783	159,783	159,783	159,783	159,783	142,761

Notes: Standard errors clustered at the household level in parentheses. Column (7) adds the Cholesky-identified government spending shock and month-of-year fixed effects; the reduction in observations reflects the sample period covered by the structural VAR. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table C-4: Effect of Spending Narrative on $E_{it}\pi_{t+12}$: Sequential Controls

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
$spending_{t-1}$	0.056*** (0.002)	0.030*** (0.002)	0.029*** (0.002)	0.035*** (0.002)	0.034*** (0.002)	0.013*** (0.002)	0.018*** (0.003)
π_{t-1}		0.240*** (0.010)	0.241*** (0.010)	0.247*** (0.010)	0.247*** (0.010)	-0.031* (0.016)	-0.060*** (0.017)
IP_{t-1}			-2.334*** (0.282)	-2.324*** (0.282)	-2.330*** (0.282)	-1.709*** (0.284)	-0.860** (0.373)
$ECB_{rate_{t-1}}$				-0.193*** (0.022)	-0.194*** (0.023)	-0.703*** (0.040)	-0.821*** (0.053)
$total_fiscal_{t-1}$					0.000 (0.000)	-0.000 (0.000)	0.001* (0.001)
ifo_klima_{t-1}						-0.039*** (0.005)	-0.061*** (0.006)
$unemp_rate_{t-1}$						-0.541*** (0.065)	-0.770*** (0.100)
$energy_idx_{t-1}$						0.055*** (0.004)	0.045*** (0.004)
$bund_yield_{t-1}$						0.077 (0.075)	0.336*** (0.098)
$fiscal_shock_{t-1}$							0.429*** (0.040)
Household FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Month-of-year FE	No	No	No	No	No	No	Yes
Observations	159,783	159,783	159,783	159,783	159,783	159,783	142,761

Notes: Standard errors clustered at the household level in parentheses. Column (7) adds the Cholesky-identified government spending shock and month-of-year fixed effects; the reduction in observations reflects the sample period covered by the structural VAR. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

D Effects of individual newspapers

Table D-5: Effect of articles implying fiscal expansion causes inflation on $E_{it}\pi_{t+12}$

	SZ	FAZ	TAZ	Bild
expansion_inflation _{t-1}	0.023*** (0.002)	-0.005 (0.004)	0.099*** (0.005)	0.085*** (0.006)
π_{t-1}	0.223*** (0.010)	0.260*** (0.010)	0.206*** (0.009)	0.239*** (0.010)
IP_{t-1}	-3.494*** (0.284)	-3.152*** (0.286)	-3.522*** (0.284)	-3.361*** (0.283)
ECB_rate_{t-1}	-0.151*** (0.023)	-0.214*** (0.023)	-0.158*** (0.023)	-0.119*** (0.023)
total_fiscal _{t-1}	0.002*** (0.000)	0.005*** (0.000)	0.004*** (0.000)	0.001*** (0.000)
Household FE	yes	yes	yes	yes
Observations	159783	159783	159783	159783
R ²	0.032	0.031	0.035	0.032

Notes: Standard errors clustered at the household level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Winsorized survey data at 2% and 98%.

Table D-6: Effect of articles implying unsustainable government debt on $E_{it}\pi_{t+12}$

	SZ	FAZ	TAZ	Bild
debt_unsustainable _{t-1}	-0.019*** (0.002)	0.020*** (0.004)	0.051*** (0.004)	0.023*** (0.003)
π_{t-1}	0.252*** (0.010)	0.263*** (0.010)	0.269*** (0.010)	0.266*** (0.010)
IP_{t-1}	-3.698*** (0.287)	-3.454*** (0.289)	-3.028*** (0.283)	-3.422*** (0.286)
ECB_rate_{t-1}	-0.201*** (0.023)	-0.206*** (0.023)	-0.183*** (0.023)	-0.197*** (0.023)
total_fiscal _{t-1}	0.006*** (0.000)	0.004*** (0.000)	0.003*** (0.000)	0.003*** (0.000)
Household FE	yes	yes	yes	yes
Observations	159783	159783	159783	159783
R ²	0.032	0.031	0.032	0.031

Notes: Standard errors clustered at the household level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Winsorized survey data at 2% and 98%.

Table D-7: Effect of articles implying the government should save more on $E_{it}\pi_{t+12}$

	SZ	FAZ	TAZ	Bild
saving _{t-1}	0.005*** (0.001)	0.009*** (0.003)	0.049*** (0.003)	0.010*** (0.002)
π_{t-1}	0.262*** (0.010)	0.261*** (0.010)	0.275*** (0.010)	0.268*** (0.010)
IP_{t-1}	-3.084*** (0.286)	-3.196*** (0.282)	-2.765*** (0.283)	-3.147*** (0.282)
ECB_rate_{t-1}	-0.206*** (0.023)	-0.212*** (0.023)	-0.171*** (0.023)	-0.194*** (0.023)
total_fiscal _{t-1}	0.003*** (0.000)	0.004*** (0.000)	0.000 (0.000)	0.002*** (0.000)
Household FE	yes	yes	yes	yes
Observations	159783	159783	159783	159783
R ²	0.031	0.031	0.032	0.031

Notes: Standard errors clustered at the household level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Winsorized survey data at 2% and 98%.

Table D-8: Effect of articles implying the government should spend more on $E_{it}\pi_{t+12}$

	SZ	FAZ	TAZ	Bild
spending _{t-1}	0.010*** (0.001)	0.009*** (0.002)	0.018*** (0.001)	0.028*** (0.002)
π_{t-1}	0.257*** (0.010)	0.263*** (0.010)	0.244*** (0.010)	0.245*** (0.010)
IP_{t-1}	-2.606*** (0.284)	-3.102*** (0.280)	-2.937*** (0.282)	-2.597*** (0.279)
ECB_rate_{t-1}	-0.207*** (0.023)	-0.202*** (0.023)	-0.220*** (0.023)	-0.195*** (0.023)
total_fiscal _{t-1}	0.001*** (0.000)	0.004*** (0.000)	0.002*** (0.000)	0.002*** (0.000)
Household FE	yes	yes	yes	yes
Observations	159783	159783	159783	159783
R ²	0.032	0.031	0.034	0.032

Notes: Standard errors clustered at the household level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Winsorized survey data at 2% and 98%.

E Non-winsorized survey data

Table E-9: Effect of articles implying fiscal expansion causes inflation on $E_{it}\pi_{t+12}$

	(1)	(2)	(3)	(4)	(5)
expansion_inflation $_{t-1}$	0.205*** (0.007)	0.107*** (0.006)	0.112*** (0.006)	0.104*** (0.006)	0.106*** (0.008)
π_{t-1}		0.209*** (0.011)	0.206*** (0.011)	0.217*** (0.011)	0.217*** (0.011)
IP_{t-1}			-3.848*** (0.350)	-3.861*** (0.350)	-3.864*** (0.350)
ECB_rate_{t-1}				-0.094*** (0.027)	-0.089*** (0.028)
total_fiscal $_{t-1}$					-0.000 (0.000)
Household FE	yes	yes	yes	yes	yes
Observations	159783	159783	159783	159783	159783
R ²	0.014	0.022	0.023	0.023	0.023

Notes: Standard errors clustered at the household level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table E-10: Effect of articles implying unsustainable government debt on $E_{it}\pi_{t+12}$

	(1)	(2)	(3)	(4)	(5)
debt_unsustainable $_{t-1}$	0.012*** (0.004)	0.021*** (0.004)	0.023*** (0.004)	0.032*** (0.004)	0.010** (0.004)
π_{t-1}		0.277*** (0.011)	0.277*** (0.011)	0.288*** (0.011)	0.267*** (0.011)
IP_{t-1}			-3.318*** (0.350)	-3.476*** (0.349)	-3.467*** (0.349)
ECB_rate_{t-1}				-0.165*** (0.027)	-0.217*** (0.027)
total_fiscal $_{t-1}$					0.004*** (0.000)
Household FE	yes	yes	yes	yes	yes
Observations	159783	159783	159783	159783	159783
R ²	0.000	0.019	0.020	0.021	0.021

Notes: Standard errors clustered at the household level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table E-11: Effect of articles implying the government should save more on $E_{it}\pi_{t+12}$

	(1)	(2)	(3)	(4)	(5)
saving _{t-1}	0.019*** (0.002)	0.013*** (0.002)	0.013*** (0.002)	0.018*** (0.001)	0.013*** (0.002)
π_{t-1}		0.273*** (0.011)	0.273*** (0.011)	0.284*** (0.011)	0.275*** (0.011)
IP_{t-1}			-3.132*** (0.348)	-3.234*** (0.348)	-3.308*** (0.348)
ECB_rate_{t-1}				-0.187*** (0.027)	-0.206*** (0.028)
total_fiscal _{t-1}					0.002*** (0.001)
Household FE	yes	yes	yes	yes	yes
Observations	159783	159783	159783	159783	159783
R ²	0.001	0.020	0.020	0.022	0.022

Notes: Standard errors clustered at the household level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table E-12: Effect of articles implying the government should spend more on $E_{it}\pi_{t+12}$

	(1)	(2)	(3)	(4)	(5)
spending _{t-1}	0.056*** (0.002)	0.029*** (0.002)	0.028*** (0.002)	0.034*** (0.002)	0.035*** (0.002)
π_{t-1}		0.244*** (0.011)	0.246*** (0.011)	0.252*** (0.011)	0.252*** (0.011)
IP_{t-1}			-2.543*** (0.346)	-2.533*** (0.346)	-2.523*** (0.345)
ECB_rate_{t-1}				-0.208*** (0.027)	-0.207*** (0.028)
total_fiscal _{t-1}					-0.000 (0.000)
Household FE	yes	yes	yes	yes	yes
Observations	159783	159783	159783	159783	159783
R ²	0.008	0.021	0.022	0.023	0.023

Notes: Standard errors clustered at the household level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

F Results non-weighted fiscal narrative indicators

Table F-13: Effect of articles implying fiscal expansion causes inflation on $E_{it}\pi_{t+12}$

	(1)	(2)	(3)	(4)	(5)
expansion_inflation $_{t-1}$	0.046*** (0.001)	0.024*** (0.001)	0.026*** (0.001)	0.024*** (0.001)	0.027*** (0.002)
π_{t-1}		0.187*** (0.010)	0.182*** (0.010)	0.191*** (0.010)	0.191*** (0.010)
IP_{t-1}			-3.932*** (0.287)	-3.924*** (0.287)	-4.020*** (0.289)
ECB_rate_{t-1}				-0.073*** (0.023)	-0.046** (0.023)
total_fiscal $_{t-1}$					-0.000*** (0.000)
Household FE	yes	yes	yes	yes	yes
Observations	159783	159783	159783	159783	159783
R ²	0.024	0.032	0.033	0.033	0.033

Notes: Standard errors clustered at the household level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Winsorized survey data at 2% and 98%. Non-weighted newspaper data.

Table F-14: Effect of articles implying unsustainable government debt on $E_{it}\pi_{t+12}$

	(1)	(2)	(3)	(4)	(5)
debt_unsustainable $_{t-1}$	0.001 (0.001)	0.003*** (0.001)	0.003*** (0.001)	0.005*** (0.001)	0.000 (0.001)
π_{t-1}		0.272*** (0.009)	0.273*** (0.009)	0.282*** (0.010)	0.261*** (0.010)
IP_{t-1}			-2.968*** (0.282)	-3.053*** (0.282)	-3.059*** (0.282)
ECB_rate_{t-1}				-0.144*** (0.023)	-0.173*** (0.023)
total_fiscal $_{t-1}$					0.001*** (0.000)
Household FE	yes	yes	yes	yes	yes
Observations	159783	159783	159783	159783	159783
R ²	0.000	0.028	0.028	0.030	0.030

Notes: Standard errors clustered at the household level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Winsorized survey data at 2% and 98%. Non-weighted newspaper data.

Table F-15: Effect of articles implying the government should save more on $E_{it}\pi_{t+12}$

	(1)	(2)	(3)	(4)	(5)
saving _{t-1}	0.007*** (0.000)	0.005*** (0.000)	0.005*** (0.000)	0.007*** (0.000)	0.008*** (0.001)
π_{t-1}		0.267*** (0.009)	0.267*** (0.009)	0.276*** (0.010)	0.287*** (0.010)
IP_{t-1}			-2.843*** (0.283)	-2.906*** (0.283)	-2.867*** (0.283)
ECB_rate_{t-1}				-0.190*** (0.023)	-0.181*** (0.023)
total_fiscal _{t-1}					-0.000*** (0.000)
Household FE	yes	yes	yes	yes	yes
Observations	159783	159783	159783	159783	159783
R ²	0.002	0.029	0.029	0.031	0.031

Notes: Standard errors clustered at the household level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Winsorized survey data at 2% and 98%. Non-weighted newspaper data.

Table F-16: Effect of articles implying the government should spend more on $E_{it}\pi_{t+12}$

	(1)	(2)	(3)	(4)	(5)
spending _{t-1}	0.010*** (0.000)	0.006*** (0.000)	0.006*** (0.000)	0.007*** (0.000)	0.009*** (0.000)
π_{t-1}		0.244*** (0.010)	0.245*** (0.010)	0.252*** (0.010)	0.262*** (0.010)
IP_{t-1}			-2.406*** (0.281)	-2.411*** (0.281)	-2.303*** (0.281)
ECB_rate_{t-1}				-0.207*** (0.022)	-0.188*** (0.022)
total_fiscal _{t-1}					-0.001*** (0.000)
Household FE	yes	yes	yes	yes	yes
Observations	159783	159783	159783	159783	159783
R ²	0.011	0.031	0.031	0.034	0.034

Notes: Standard errors clustered at the household level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Winsorized survey data at 2% and 98%. Non-weighted newspaper data.

G Composite indicator including all four narratives

As a robustness check, we construct a composite fiscal narrative index that includes all four narrative indicators. The spending and unsustainable debt narratives are weighted positively, as both show amplifying effects on fiscal shock transmission. The saving narrative is weighted negatively (multiplied by -1), as it dampens transmission. The expansion→inflation narrative is weighted positively, though its transmission effect is less pronounced than the spending narrative. The resulting index is standardized to unit variance.

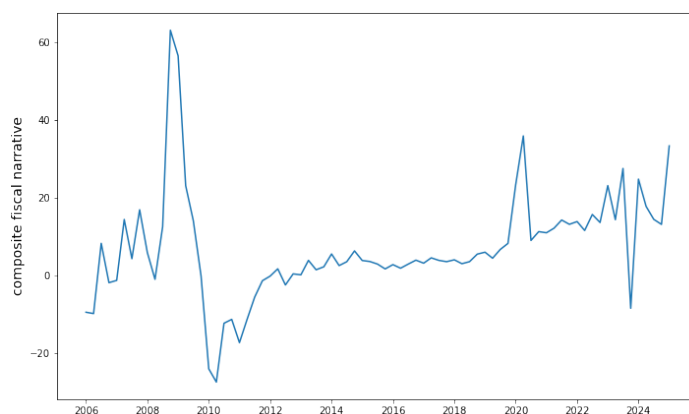


Figure G-4: Standardized composite fiscal narrative index constructed from all four narrative indicators. The spending and unsustainable debt narratives enter with a positive weight, the saving narrative enters with a negative weight (multiplied by -1), and the expansion→inflation narrative enters with a positive weight.

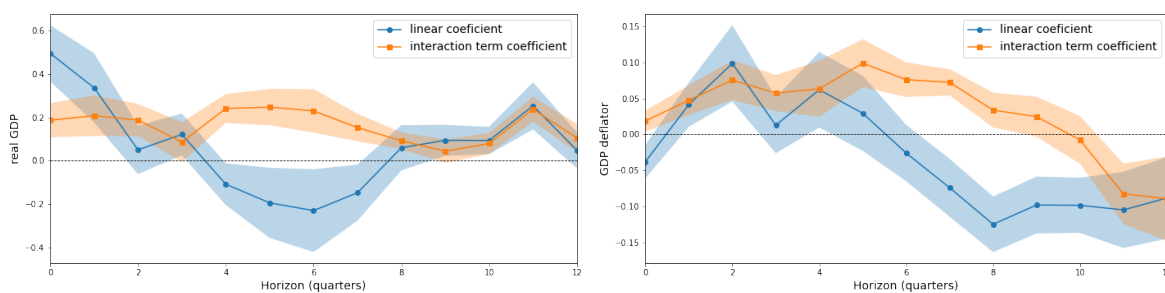


Figure G-5: This figure presents the impulse responses of a government spending shock (blue) on GDP (left panel) and the GDP deflator (right panel) and its interaction with the composite fiscal narrative indicator including all four narratives (orange) together with 68% confidence intervals.